

For Online Publication

A Appendix Tables

Table A.1: Construction of 701 Adopter and Main Analysis Samples

Sample Restriction	Total Munis	701 Adopters	Adoption Rate
All zoning jurisdictions in CBSAs	20,956	3,093	14.8%
Municipalities with historical population	15,222	2,968	19.5%
+ between [1000, 50000] 1960 residents	7,441	2,579	34.7%
Balanced panel of municipalities since 1970	2,561	1,264	49.4%
+ distance to CBD filter	2,153	1,081	50.2%

Notes: This table overviews the data processing steps we take to process the universe of U.S. zoning jurisdictions into the samples we use for 701 assistance statistics and regression analysis. We report the number of adopters at each successive sample restriction, based on if they are matched in our 701 project directories, as well as total municipalities. The balanced panel of municipalities includes municipalities above 50,000 residents that are ineligible for 701 assistance.

Table A.2: Overall covariate means between adopters and non-adopters

Variable	Mean Values		Difference	P-value
	Treated	Control		
Distance to metro CBD (km)	29305.633	28987.201	318.432	0.624
Population Growth Rate 1950-1960	0.537	0.479	0.058	0.115
Highway Intersects Municipality	0.318	0.201	0.116***	0.000
Distance to Highway (km)	17.396	19.206	-1.810**	0.023
Number of Households (1940)	1815.153	1273.490	541.663***	0.000
Total Income per Household (1940 \$)	1516.372	1461.267	55.105***	0.000
Average Home Value (1940 \$)	3445.462	3116.748	328.713***	0.000
College Graduate Share	0.093	0.084	0.009***	0.000
Blue Collar Worker Share	0.442	0.427	0.016***	0.000
Agricultural Worker Share	0.140	0.184	-0.044***	0.000
Manufacturing Worker Share	0.209	0.204	0.005*	0.068

Significance levels: * = 10%; ** = 5%; *** = 1%.

Notes: Following the tabulation of 1940 covariates as described in Section 4.1, this table compares means tabulated based on 701 adoption status, as recorded in the HUD/HHFA project directories. Municipalities analyzed have 1960 Census populations of at least 1,000 and at most 50,000; ineligible municipalities are excluded in this Table. Units of the variable are reported in the leftmost column. All covariates tested in this table are also used in the covariate balance tests of Table 1.

Table A.3: Primary outcome results using 701 adoption treatment

	(1)	(2)	(3)	(4)	(5)	(6)
Outcomes - One side restricted ($\leq 200K$)						
	(Log) New Units	% Apt Units (Cu- mul. to 2010)	% SF Units (Cu- mul. to 2010)	(Log) Med. Home Values	Moderate Owned Unit Shares	MLS Requirement Restrictiveness
Coefficient	0.153*** (0.038)	0.850** (0.367)	-1.137** (0.451)	0.087*** (0.019)	-2.926*** (1.044)	1.524*** (0.266)
Observations	15,659	15,659	15,659	13,993	13,056	17,806
R ² (within)	0.038	0.050	0.025	0.028	0.050	0.016
Outcomes - Restricted (5K – 200K)						
	(Log) New Units	% Apt Units (Cu- mul. to 2010)	% SF Units (Cu- mul. to 2010)	(Log) Med. Home Values	Moderate Owned Unit Shares	MLS Requirement Restrictiveness
Coefficient	0.263*** (0.043)	0.312 (0.435)	-0.697 (0.485)	0.112*** (0.021)	-5.432*** (1.173)	1.577*** (0.314)
Observations	10,654	10,654	10,654	9,746	8,906	12,145
R ² (within)	0.046	0.037	0.023	0.036	0.055	0.019

Significance levels: * = 10%; ** = 5%; *** = 1%.

Notes: This table reports estimates using confirmed 701 adoption as the treatment variable rather than eligibility interacted with state adoption exposure. Columns report the outcome listed in the column heading. Coefficients are from variants of the triple-difference specification estimated on the samples indicated in the panel headings. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; Cui (2024) lot-size restrictiveness measures; and transportation access controls from Atack (2017) and Weiwu (2024).

Table A.4: Primary outcome results accounting for heterogeneous spillover effects

	(1)	(2)	(3)	(4)	(5)	(6)
	Outcomes - Restricted (5K – 200K)					
	(Log) New Units	% Apt Units (Cu- mul. to 2010)	% SF Units (Cu- mul. to 2010)	(Log) Med. Home Values	Moderate Owned Unit Shares	MLS Requirement Restrictiveness
<i>Panel A: Coefficients for adopted and for eligible non-adoptees</i>						
β_{adopt}	-0.649** (0.326)	-12.898*** (3.669)	10.617*** (3.781)	0.342** (0.143)	-12.996 (7.982)	4.929** (2.218)
β_{elig}	-1.113*** (0.374)	-6.617 (4.140)	11.702*** (4.372)	0.215 (0.153)	-22.389** (9.333)	1.919 (2.608)
<i>Panel B: Comparison of identified treatment effects on the treated</i>						
Original τ_{treat}	-0.200	-2.818	2.928	0.085	-4.877	1.286
Het. τ_{treat}	-0.182	-3.614	2.975	0.096	-3.642	1.381
Combined τ_{elig}	-0.204	-2.683	2.765	0.075	-4.102	0.971
Observations	10,654	10,654	10,654	9,746	8,906	12,145
R ² (het. model)	0.057	0.050	0.031	0.080	0.074	0.025

Significance levels: * = 10%; ** = 5%; *** = 1%.

Notes: This table reports estimates from a specification that allows effects to differ between confirmed 701 adopters and eligible non-adopting municipalities. Panel A reports separate exposure coefficients for confirmed adopters and eligible non-adopters. Panel B compares the implied treatment effects from the baseline and heterogeneous-effect specifications. The sample is restricted to municipalities with 1960 populations between 5,000 and 200,000. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; Cui (2024) lot-size restrictiveness measures.

Table A.5: Persistent associations of MLS restrictiveness with market outcomes

Outcome		Pre-1960s Variation		Post-1960s Variation		
		1 Lag	2 Lags	0 Lags	1 Lag	2 Lags
Excess Land	Coefficient	0.555** (0.191)	0.390** (0.153)	—	0.665*** (0.028)	0.497*** (0.045)
	R ² (within)	0.229	0.104		0.531	0.370
% Apartments	Coefficient	-0.220** (0.076)	-0.234** (0.089)	-0.338*** (0.030)	-0.273*** (0.034)	-0.187*** (0.038)
	R ² (within)	0.010	0.006	0.029	0.020	0.012
% Single Family	Coefficient	0.466** (0.175)	0.430** (0.131)	0.593*** (0.049)	0.517*** (0.059)	0.439*** (0.072)
	R ² (within)	0.032	0.022	0.071	0.059	0.046
Log(Price)	Coefficient	0.000 (0.004)	0.002 (0.003)	0.008*** (0.001)	0.009*** (0.001)	0.010*** (0.002)
	R ² (within)	0.012	0.020	0.041	0.045	0.050
Moderate Segment Share	Coefficient	-0.121 (0.250)	-0.056 (0.192)	-0.297*** (0.052)	-0.338*** (0.051)	-0.479*** (0.046)
	R ² (within)	0.010	0.009	0.017	0.020	0.037
Bottom Segment Share	Coefficient	-0.131 (0.198)	-0.081 (0.123)	-0.263*** (0.047)	-0.285*** (0.058)	-0.395*** (0.042)
	R ² (within)	0.028	0.022	0.031	0.030	0.047
Not Cost Burdened Renters	Coefficient	0.050 (0.237)	0.012 (0.193)	-0.215*** (0.045)	-0.234*** (0.059)	-0.244*** (0.054)
	R ² (within)	0.051	0.048	0.046	0.038	0.035
Ownership rates	Coefficient	0.280** (0.099)	0.271*** (0.065)	0.341*** (0.039)	0.351*** (0.040)	0.367*** (0.048)
	R ² (within)	0.048	0.042	0.069	0.074	0.084

Significance levels: * = 10%; ** = 5%; *** = 1%.

Notes: This table reports associations between lot-size restrictiveness and housing-market outcomes over time. Each coefficient comes from a regression of the row outcome on contemporaneous or lagged MLS restrictiveness, with columns separating pre- and post-1960s variation. Specifications include Census-region-by-decade fixed effects and baseline controls. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); Cui (2024) lot-size restrictiveness measures; and transportation access controls from Atack (2017) and Weiwu (2024).

Table A.6: Table 2 results under different bandwidths

	(1)	(2)	(3)	(4)	(5)
(Log) New Units - Full sample					
Coefficient	-0.772*** (0.280)	-0.658** (0.286)	-0.822*** (0.271)	-0.838*** (0.270)	-0.810*** (0.252)
Implied ATT	-0.228	-0.194	-0.242	-0.247	-0.239
Observations	17,024	16,065	16,058	16,058	16,058
R ² (within)	0.035	0.112	0.069	0.065	0.062
(Log) New Units - More Restricted (5k–150k)					
Coefficient	-0.658* (0.344)	-0.598* (0.350)	-0.593* (0.332)	-0.635* (0.332)	-0.565* (0.307)
Implied ATT	-0.194	-0.176	-0.175	-0.187	-0.166
Observations	11,067	10,535	10,528	10,528	10,528
R ² (within)	0.020	0.052	0.056	0.051	0.047
(Log) New Units - Donut hole around 50k					
Coefficient	-0.656** (0.332)	-0.668** (0.324)	-0.649** (0.317)	-0.724** (0.319)	-0.646** (0.298)
Implied ATT	-0.193	-0.197	-0.191	-0.213	-0.190
Observations	10,248	9,751	9,744	9,744	9,744
R ² (within)	0.018	0.053	0.052	0.047	0.044
Pop. Controls		✓	✓	✓	✓
Location Controls			✓	✓	✓
Region × Year FE				✓	
Division × Year FE					✓

Significance levels: * = 10%; ** = 5%; *** = 1%.

Notes: This table reports robustness of the housing-supply estimates in Table 2 to alternative population bandwidths around the 50,000-person eligibility threshold. Across columns, specifications vary in controls and fixed effects as indicated. Across panels, the sample changes from the full one-sided sample to narrower and donut-hole restrictions. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; and transportation access controls from Atack (2017) and Weiwu (2024).

Table A.7: Primary outcome results over placebo eligibility threshold

	(1)	(2)	(3)	(4)	(5)	(6)	
	Outcomes - Restricted w/ 20K threshold						
	(Log) New Units	% Units (Cumul. 2010)	Apt (Cu- to 2010)	% SF Units (Cumul. to 2010)	(Log) Med. Home Val- ues	Moderate Owned Unit Shares	MLS Re- quirement Restrictive- ness
Coefficient	-0.107 (0.297)	-0.633 (3.064)	2.934 (3.488)	0.133 (0.144)	-4.400 (7.292)	0.181 (2.364)	
Observations	9,121	9,121	9,121	8,080	7,603	10,393	
R ² (within)	0.059	0.066	0.034	0.081	0.083	0.026	

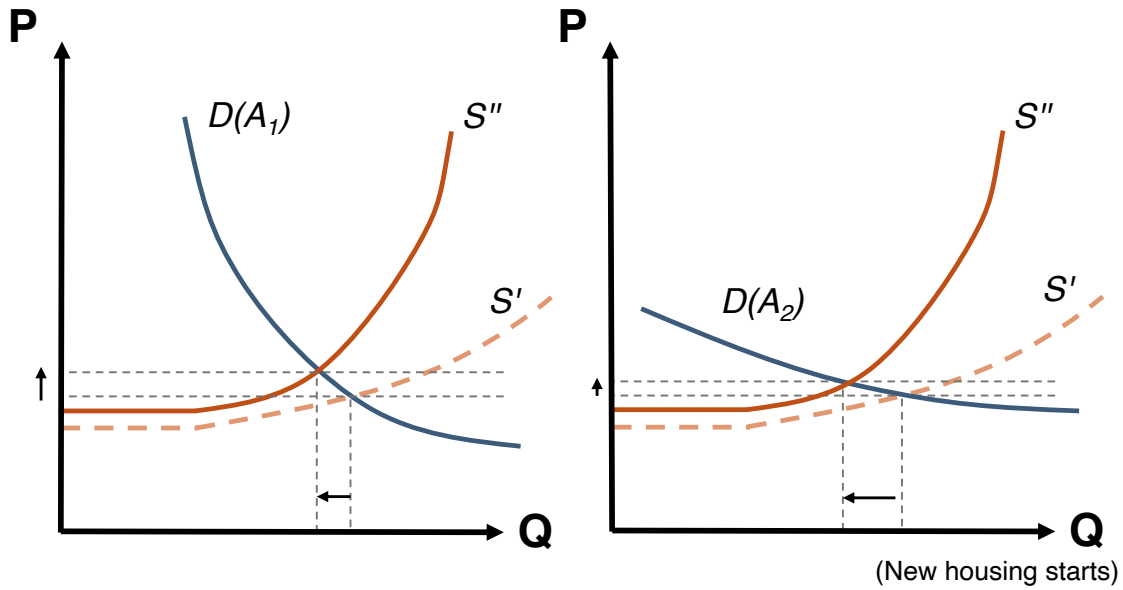
Significance levels: * = 10%; ** = 5%; *** = 1%.

Notes: This table reports placebo-threshold estimates that redefine 701 eligibility using a 20,000-person population cutoff among municipalities below the true 50,000-person threshold. Columns report the outcome listed in the column heading. Specifications otherwise follow the primary triple-difference design in Equation 2. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; Cui (2024) lot-size restrictiveness measures.

B Appendix Exhibits

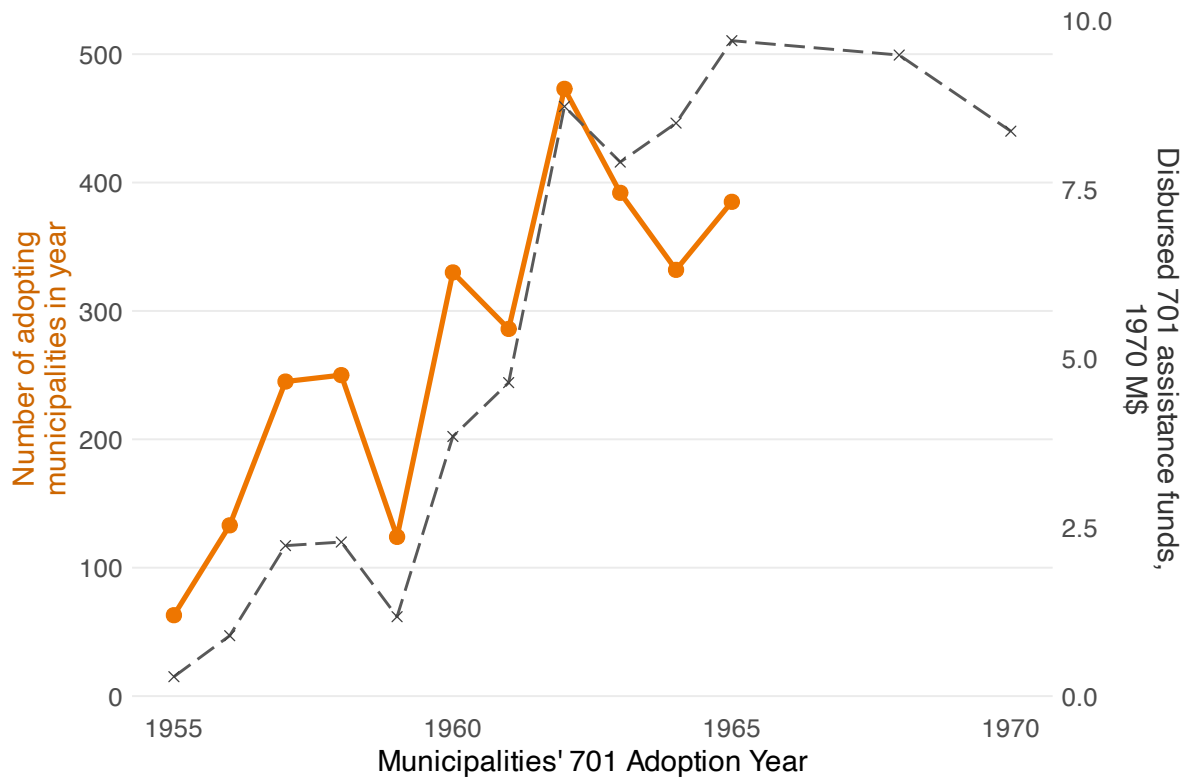
Figure B.1: Housing market outcomes after local growth management



Notes: This schematic illustrates the housing-market mechanism described in Section 2.3. Growth-management regulation is represented as an inward shift in the supply of new housing from S' to S'' . Holding demand fixed, the framework predicts fewer new units and higher housing prices; the magnitude of price changes depends on local demand elasticity and on substitution across municipalities and housing segments.

Source: Authors' illustration.

Figure B.2: The scaling up of 701 grants and expenditures

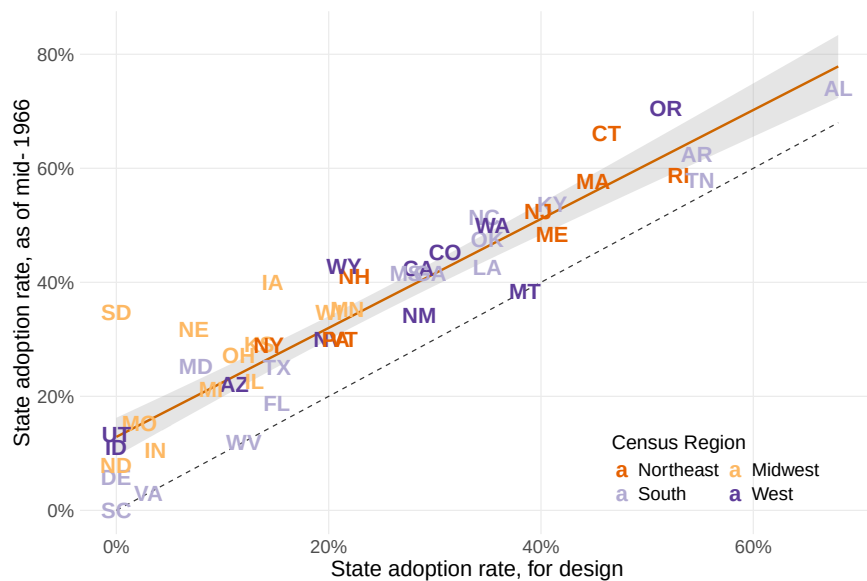


Notes: This figure plots the annual scale-up of Section 701 assistance in the project-directory data. The series report newly approved municipalities and federal grant amounts over time. Grant amounts are measured using the amounts recorded in the HUD/HHFA project directories.

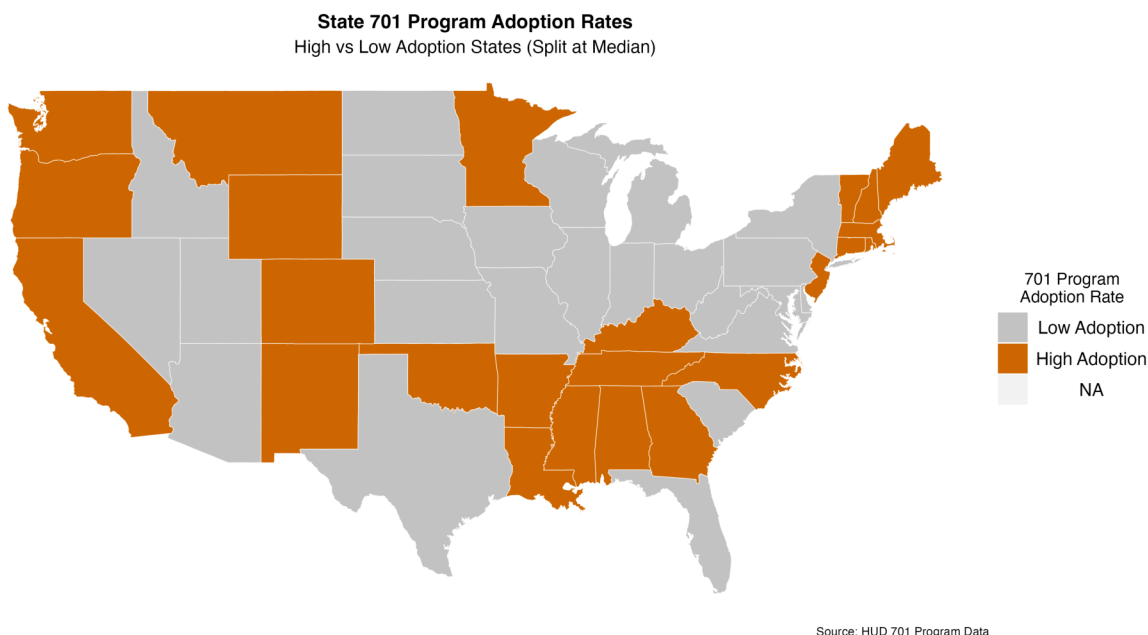
Sources: HUD/HHFA 701 project directories; Schmidt (2018) population panel data.

Figure B.3: Further state-level trends in adoption rates

(a) Scatterplot of adoption variation over time



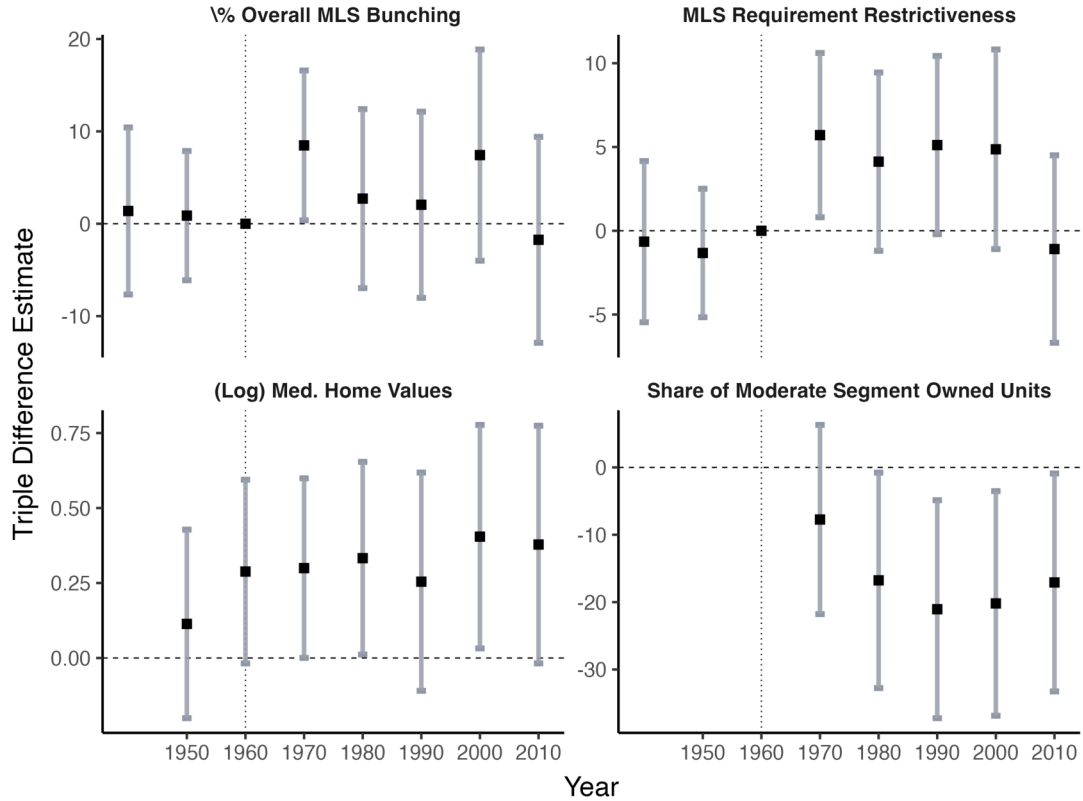
(b) Two tiers of state-level 701 adoption rates, as of 1962



Notes: Both panels plot variation in the state adoption rate variable $\tilde{Z}_s$, which we use as a measure of 701 policy exposure. Panel (a) separates the states at the threshold of $\tilde{Z}_s = 23\%$, which is the median value of state adoption rates computed over the municipalities in our national sample. In Panel (b), we plot the correlation between adoption rate constructions defined as of 1962 and as of the latest project directory available in 1966.

Figure B.4: Triple difference event studies for non-supply outcomes

Land Use and Price Effects

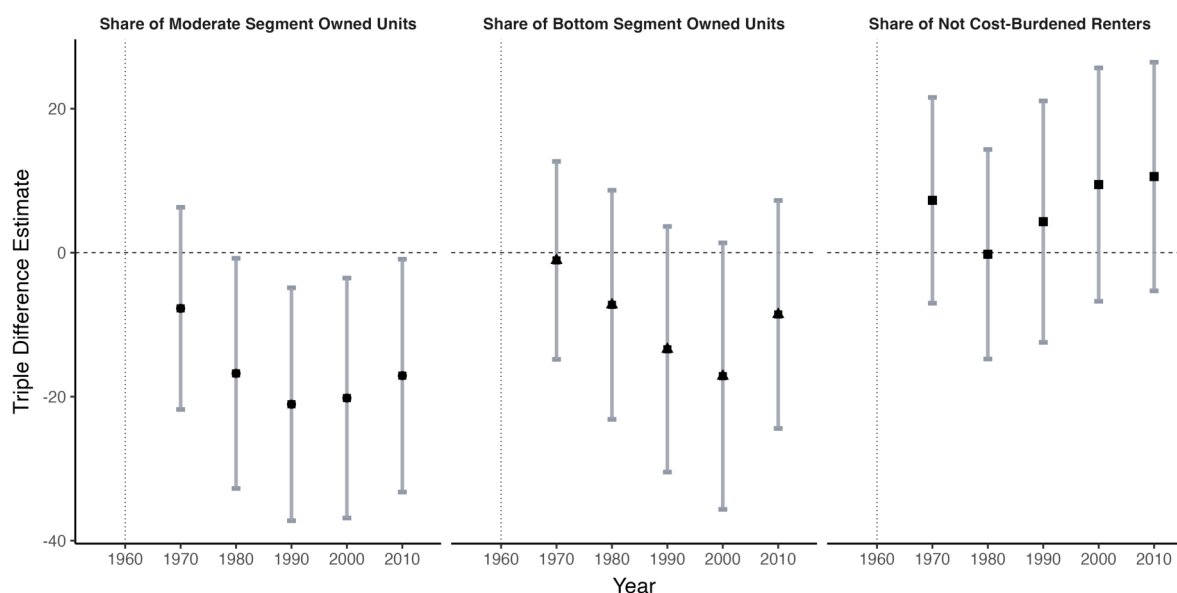


Notes: This figure plots coefficients from the event-study version of the triple-difference design in Equation 4 for non-supply outcomes. Outcomes include lot-size restrictiveness, large-lot restrictiveness, log median owner-occupied home value, and the share of owner-occupied units below the CBSA median home value. Estimates use the restricted sample of municipalities with 1960 populations between 5,000 and 200,000. Points report event-time coefficients; bars report 95% confidence intervals. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; Cui (2024) lot-size restrictiveness measures; and transportation access controls from Attack (2017) and Weiwu (2024).

Figure B.5: Triple difference event studies for affordability across market segments

Affordability Effects: 701 Program Event Study

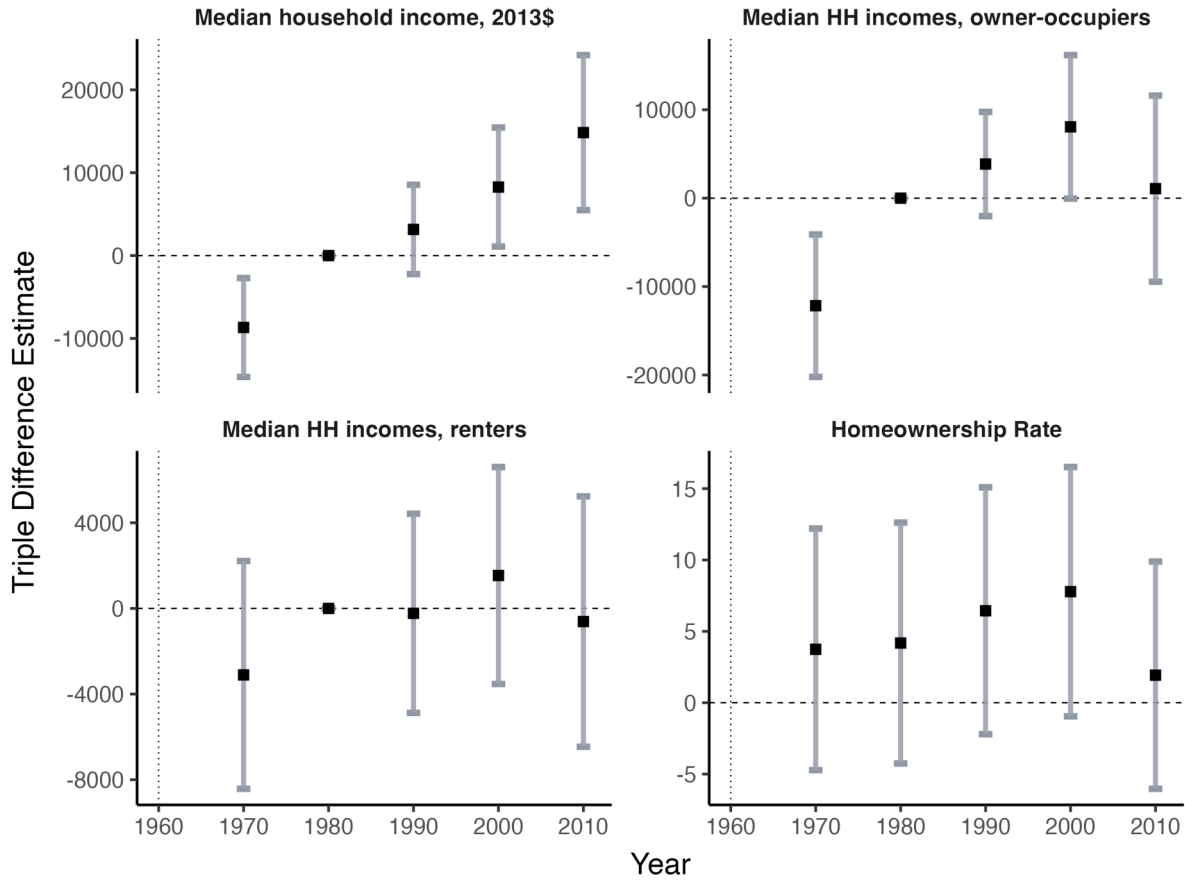


Notes: This figure plots coefficients from the event-study version of the triple-difference design in Equation 4 for affordability outcomes across market segments. Estimates use the restricted sample of municipalities with 1960 populations between 5,000 and 200,000. Points report event-time coefficients; bars report 95% confidence intervals. Standard errors are clustered at the municipality level. Further details on affordability outcome construction are in Section 3.2.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; and transportation access controls from Atack (2017) and Weiwu (2024).

Figure B.6: Triple difference event studies for median income outcomes

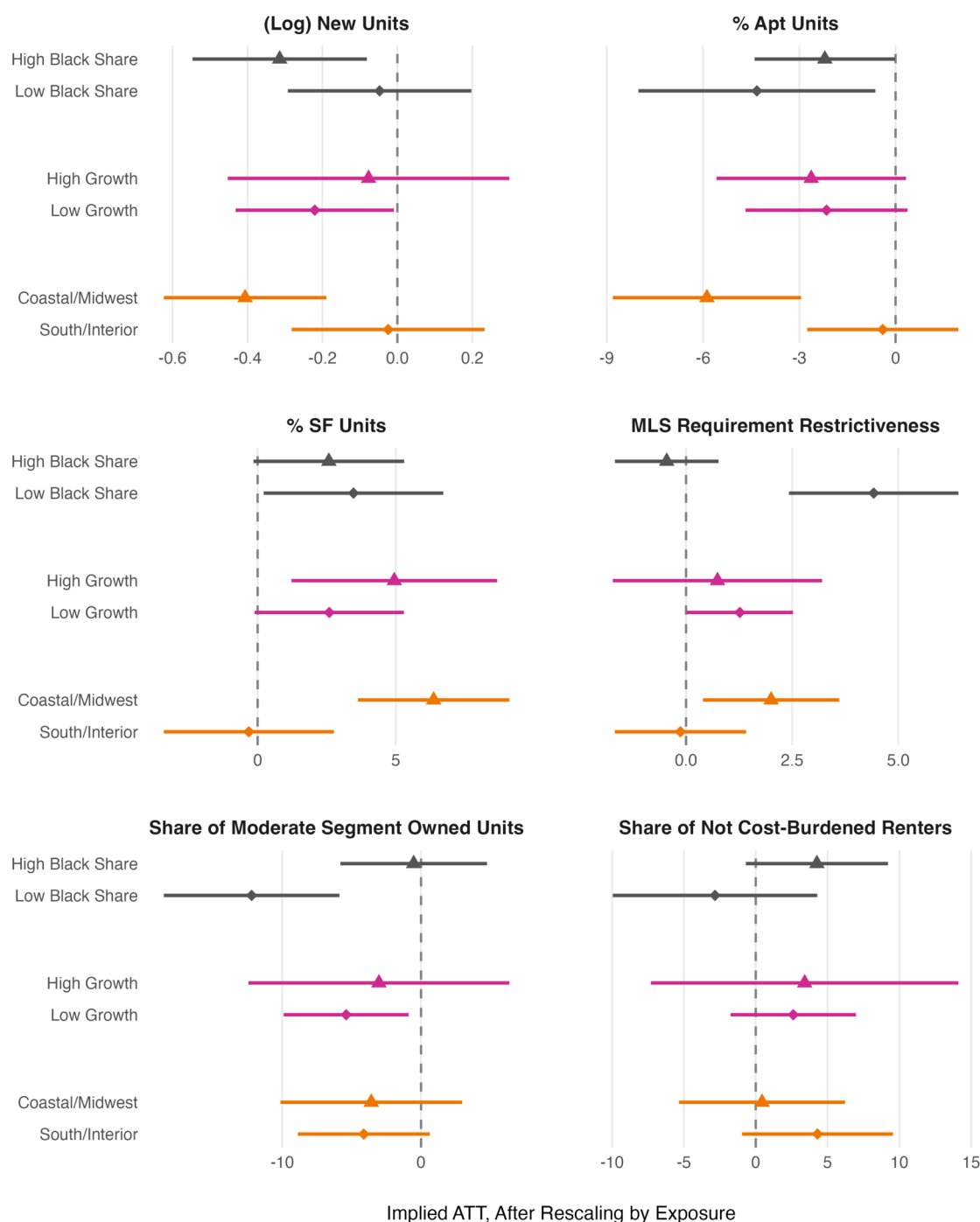
Income sorting between municipalities



Notes: This figure plots coefficients from the event-study version of the triple-difference design in Equation 4 for household income and ownership outcomes. Estimates use the restricted sample of municipalities with 1960 populations between 5,000 and 200,000. Points report event-time coefficients; bars report 95% confidence intervals. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; and transportation access controls from Attack (2017) and Weiwu (2024).

Figure B.7: Heterogeneity in supply & affordability ATTs following 701 assistance

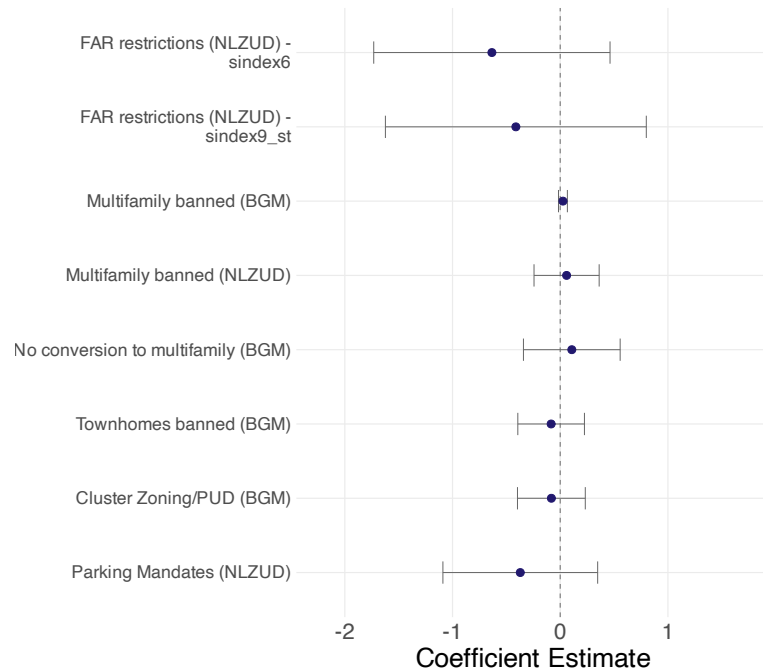


Notes: This figure plots implied average treatment effects on the treated from split-sample variants of the triple-difference specification in Equation 2. Subgroups split municipalities by baseline Black share, 1950–1960 population growth, and region. Points report exposure-rescaled treatment effects; bars report 95% confidence intervals. Standard errors are clustered at the municipality level.

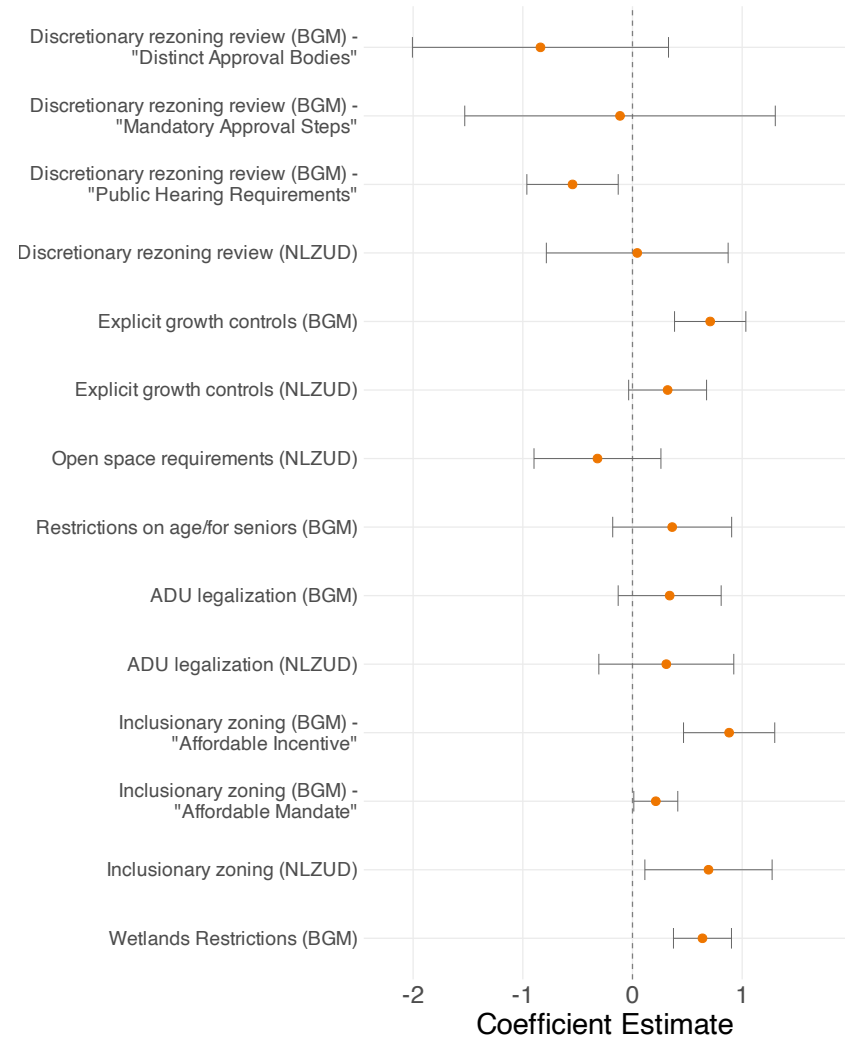
Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; and transportation access controls from Atack (2017) and Weiwu (2024).

Figure B.8: Path Dependence In Specific Growth Controls and Land Use Regulations

(a) Changes to regulations generally adopted pre-701



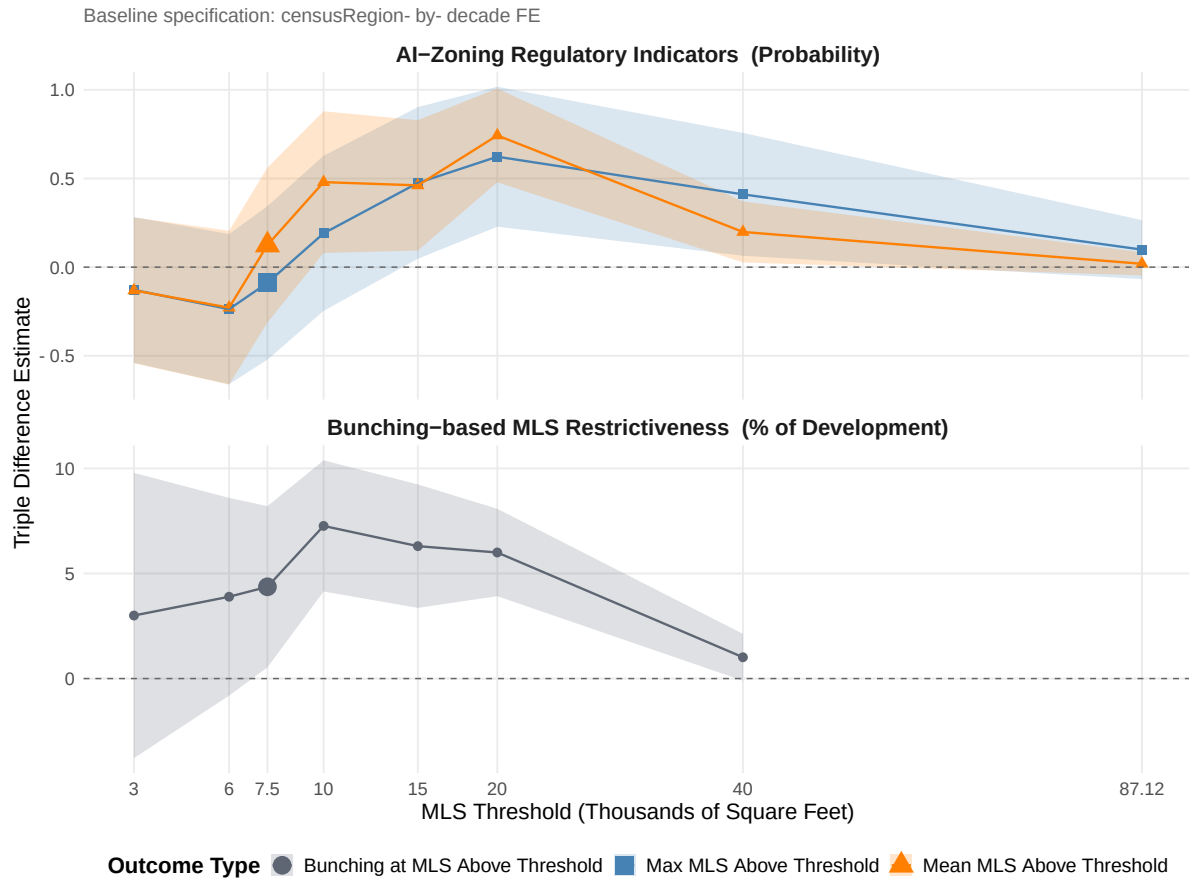
(b) Regulations generally adopted post-701



Notes: This figure plots coefficients from the cross-sectional exposure regression in Equation 5, estimated separately for individual land-use regulations recorded in the AI-Zoning and NZLUD databases. Panel (a) reports regulations generally adopted before or during the 701 program period; Panel (b) reports regulations popularized after the 1970s. Coefficients represent the effect of a one-unit increase in the state 701 adoption-rate exposure measure. 95% confidence intervals are shown. Further details are in Section 6.1.

Sources: AI-Zoning database (Bartik, Gupta and Milo, 2024); National Zoning and Land Use Database (Mlecenko and Desmond, 2023); HUD/HHFA 701 project directories; Schmidt (2018) population panel data.

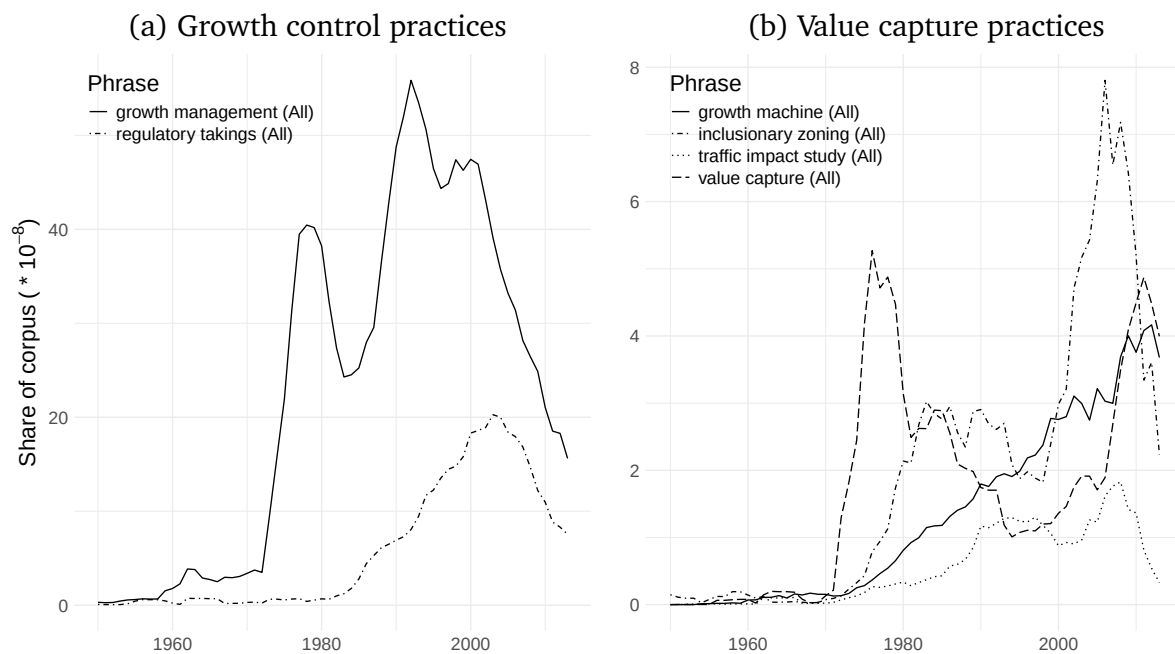
Figure B.9: Consistency of MLS restrictiveness results across measures



Notes: This figure reports triple-difference estimates for alternative minimum-lot-size restrictiveness measures across lot-size thresholds. The bunching-based panel uses the panel measure from Cui (2024); the AI-Zoning panel uses cross-sectional indicators from Bartik, Gupta and Milo (2024). Points report coefficient estimates and shaded bands report 95% confidence intervals. The larger marker denotes the baseline 7,500-square-foot threshold used elsewhere in the paper.

Sources: HUD/HHFA 701 project directories; Schmidt (2018) population panel data; Cui (2024) lot-size restrictiveness measures; and Bartik, Gupta and Milo (2024) AI-Zoning database.

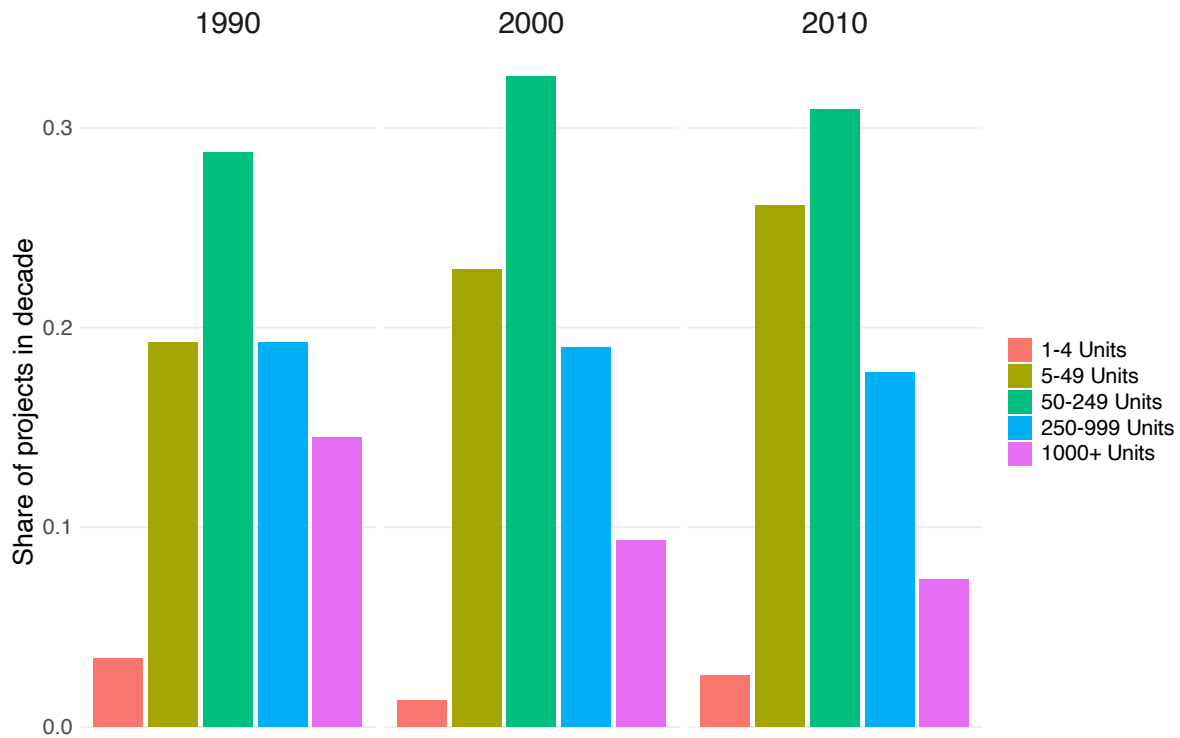
Figure B.10: Google Ngram analysis of planning practices



Notes: This figure plots Google Books Ngram frequencies for terms associated with growth-control and value-capture planning practices. Panel (a) reports terms related to growth controls; Panel (b) reports terms related to value capture. The figure is descriptive and is used to document the timing with which these planning concepts entered published discourse.

Source: Google Books Ngram corpus.

Figure B.11: Development-level distribution of unit counts



Notes: This figure reports the distribution of development unit counts for newspaper-covered developments requiring discretionary approval. Unit-count bins are coded by an LLM from the project prompt described in Appendix E; the plotted sample is restricted to developments whose municipality can be matched to the zoning-jurisdiction directory. Panels split developments by decade.

Sources: Newsbank/Access World News database; authors' LLM coding of development articles.

C Details on additional data cleaning

C.1 Classification of HUD index

After loading in the 62,314 separate titles recorded in the HUD document index, we first clean and standardize the “geographic place names” (GPNs) assigned to every title. Planning documents produced under the Section 701 program may both refer to municipalities (the focus of the small areas urban planning assistance program), or other geographies like planning areas, counties and metropolitan areas. From the roughly 15,000 unique GPNs in the index, we conduct a fuzzy string match of GPNs to a directory of zoning jurisdictions.

After the match and filtering only to the incorporated municipalities and townships in our analysis sample, we retain approximately 6,000 municipalities. Around 2,800 of the remaining municipalities were also found in the HUD/HHFA project directories for 701 planning assistance (i.e. the link to the project directory municipalities is not perfect, but has high coverage).

Over the municipalities linked to both the index file and the project directories, we classify documents associated to each municipality by keywords occurring in the document titles. A flexible keyword to capture if a document is related to zoning, for example, not only checks for the string ‘ZONING ORDINANCE’ but also related terms like ‘ZONING RESOLUTION’, ‘ZONING MAP’ and so forth. Below, we list the exact regular expressions used as the keyword to classify titles.

- Development plan: “(COMPREHENSIVE|MASTER|GENERAL|LAND USE) . *PLAN”
- Capital budget: “(CAPITAL|PUBLIC) IMP”
- Zoning-related: “ZONING (ORDI|REGUL|RESOLUT|STANDARD|MAP)”
- Subdivision regulation-related: “SUBDIVISION ”
- Further land use: “^LAND USE”
- Floodplain-related: “FLOOD |WATERSHED | STORM ”
- General environmental: “ENVIRONME”
- Open space-related: “OPEN SPACE |RECREATION’”

We allow for the same document title to be classified into multiple groups, though the deliberate usage of spaces is aimed at limiting misclassification based on the keyword matching to parts of longer words. Then, as long as a document linked to a municipality is classified to be in one of the eight types, the municipality is classified to have that type present, which is the outcome we illustrate in Figure 1.

C.2 Imputation of post-1966 adopters

Figure 1 suggests that municipalities receiving planning assistance under the 701 program often received a standard package of planning products: a document that comprises the final

plan, plus regulatory amendments related to zoning and to land use. To impute additional municipalities that may have received 701 grants after July 1966, we search for cities with both types of documents in the HUD document index.

For GPNs matched to a standardized zoning jurisdiction but not also matched to the project directory, we filter to municipalities whose earliest document in the index is 1966 or later and whose 1960 population is below 50,000 residents. We search for documents produced up to a final year, which we set to be 1973.

A municipality is imputed to be a post-1966 adopter if, among products produced for that municipality up to a final year, one of two conditions are satisfied:

1. It has one document classified as either a development plan or a capital budget, *plus* a document classified as related to zoning, subdivision or further land use;
2. There are four or more documents produced for the municipality, at least two of which are classified to be in the development plan/capital budget category or the zoning/land use category.

The filters we use represent a tradeoff that is biased towards a conservative definition of post-1966 adopter. We aim to avoid false positives, where a document may have been issued for a zoning jurisdiction which was separate from a comprehensive planning grant issued to that municipality. That is a possibility when 701 grants became more commonly issued to regional planning or state planning agencies in the late 1960s.

In addition, we use a rule of thumb that the planning products produced in the 701 process were dated two years before grants were approved and the municipalities adopted 701, using information from federal reports that the process usually took 18 months. By comparing approval dates for pre-1966 municipalities to HUD document years, we found that development plans were issued 3 years after grant approval on average. The minimum of the document years, shifted forward following the rules above, is the adoption year of post-1966 imputed municipalities.

We detect 893 additional jurisdictions through this process. In our analysis, we pool these jurisdictions along with adopters as of 1966 specifically in the spillovers specifications estimated in Table A.4. Figure C.1 superimposes the imputed time series of municipalities onto Appendix Figure B.2. While the level of 701 funds disbursed stays steady past 1966, the amount becomes more diverted away from municipalities that had been taking up federal planning assistance; the trend of imputed adopters is on a steady decline, until it reaches fewer than 100 municipalities in a year by 1970.

C.3 Details and formulas for non-supply outcomes

Construction of affordability measures. For median home values at the metropolitan level, as well for municipalities in 1970 (when medians of the distribution were not precisely reported), we calculate the median value from the (possibly aggregated) home value bin data. We use the binned home value distribution to interpolate the median value. For the first bin where 50% of units have values below the bin, the interpolated value is the midpoint of the bin's range. We also interpolate median values the same way with 1940 pre-701 data, as home values in the full count Census are reported, but rounded to the nearest hundred.

Figure C.1: Visualizing imputed trends in 701 grants



Notes: This figure extends the project-directory time series by adding post-1966 municipalities imputed from the National Archives index of HUD Section 701 final reports. Imputed adopters are jurisdictions whose indexed planning products satisfy the document-combination rules described in Appendix C. Imputed adoption years are based on the timing rule described in the text.

Sources: HUD/National Archives Section 701 final-report index; HUD/HHFA 701 project directories.

Our primary measure of affordability is to take the MSA-level median home value, then count the share of units in each municipality below the threshold. We repeat the interpolation process for the 25th quantile for each MSA, constructing a separate measure of “bottom segment” owned units that are present in each municipality.

We also construct a measure of rental affordability using MSA-level data as the denominator. For each MSA, we calculate the median renter household income at that level. Using standard definitions of rental cost burden, we define an MSA-wide affordable rent as 30% of that median. Then, using binned tabulations for rent at the municipality level, we can calculate the share of units available in the municipality whose rents would not impose a cost burden on the median renter in the wider housing market.

Segregation index for multifamily apartments. In analysis of racial segregation, it is common to use a dissimilarity index of segregation when only two racial groups are considered. We can adapt the same index by splitting the housing stock built up to a year as either units in multifamily apartments or non-apartment units.

Define the probability that a unit is in block group k , given the unit is in a municipality and

belongs to group g , as $P_{k|g}$. The dissimilarity index of segregation for a municipality is then:

$$I = \frac{1}{2} \sum_k |P_{k|A} - P_{k|B}|$$

In other words, it is the difference of ratios in how much the share of the overall group is in each block group. Using 1990 Census data, we match the municipalities in our analysis sample to block groups that are mainly inscribed in one municipality's borders. We iterate, over those block groups, the dissimilarity index calculation for each municipality as of 1990.

Bunching measure of minimum lot size restrictiveness. Though details are provided in Cui (2024), the general idea behind estimating a national panel of MLS requirements is to simultaneously back out which requirements were ever adopted, then estimate the degree of bunching at the requirement levels at any one time. The estimation is done with reference to the lot size distributions of single-family homes in a jurisdiction every decade, so it does not require any spatial information on zoning district boundaries.

We classify a lot size as being one mandated by a requirement somewhere in the municipality if we detect a concentration of development at that specific lot size, such that the density estimate at the size is much larger than the density estimate over marginally smaller lots. The excess mass measure of MLS restrictiveness requires estimating a counterfactual density without bunching induced by the lots, which we can do using kernel estimation methods leveraging density info for lots marginally larger or smaller than the lot size mandated by MLS requirements.

Unlike the Census data before it, the restrictiveness outcomes can be missing for municipalities that either did not build meaningful amounts of housing units over a decade or lack an online version of their zoning ordinances.¹ We do not impute these measures where they are missing, but we note that the median population of eligible municipalities missing lot size restrictiveness data is 624.

¹This is a separate case from, e.g., lot size restrictiveness having a value of 0, indicating that there were enough units built but no discernible bunching on lot sizes.

D More on identification assumptions

D.1 Overview of state agency practices around 701 assistance

We offer evidence from primary sources on two claims: the lack of consistency in how agencies encouraged municipalities to apply for funds, and the importance of understaffed agencies in affecting their share of 701 funds. We evaluated policy memos issued by federal agencies (HUD and the Housing and Home Finance Agency, the Department's predecessor) and documents from 17 state agencies as they approved 701 assistance funds in the 1960s.

Agencies had discretion to publicize the 701 assistance program to any municipalities in the state, as well as consult with local government on how to prepare an application for the federal grants. While we found evidence that federal officials would occasionally cap how much funding could be allocated to one state, there were no minimum allocations to states or attempts to prioritize grants for large states with low adoption.

There were examples of states at both extremes of publicity and consultation. One model used by states like Alabama, Rhode Island or Oregon was to match municipalities with planners directly employed by the agency, replacing the role of planner consultants in the process entirely. This model was also complemented by state financial assistance to make up for the gap in federal funding, leading to minimal financial and transaction costs for any municipalities interested in planning assistance.

On the other hand, New Hampshire and Virginia were particularly selective in screening applications for 701 assistance from municipalities. To pursue assistance only for localities with "concrete commitments to planning," they screened out applications from municipalities with inexperienced planning commissions or where no full-time planner was already hired (Hammer Greene Siler Associates, 1969). In another case, certain states did not pass legislation allowing for regional planning or to enable receiving federal planning funds, as also noted in Collins and Shester (2013). Idaho and Utah did not pass enabling legislation until 1965, while South Carolina did not pass all required legislation until 1968.

Our second point is that insufficient staffing of the agency caused delays in approving grants before the year's Congressional allocation for 701 funds was exhausted, which meaningfully altered the state's share of allocated 701 funds. For five states, we can retrieve the number of planners responsible not just for filing 701 applications at different points in time, but also for supervising the caseloads for grants in progress until they were completed. That allows us to calculate the average caseload per supervising planner, derived from the jurisdictions with a 701 grant approved but still not yet completed at the point in time.

In Table D.1, we report the point-in-time estimates of planner caseloads for available states. There is large variation in caseloads: active municipalities supervised by planners range from 30 to 3 municipalities, depending on the state. This measure of planner caseloads is also an underestimate of applications that supervisory planners are preparing for submission to new grants. Even when caseload ratios are similar, a state like Texas is dealing with much larger municipalities on average, so each planner is managing planning services that cover a greater number of residents.

A larger caseload per planner not only plausibly delays the time from when a new municipality begins the 701 application process to when a planner is first paid for work, but also affects whether the municipality receives 701 assistance at all. First, Hammer Greene

Table D.1: Caseload burden variation between 701 state agencies

State	Date of data	Planners	Active 701 munis	Caseload/Planner	1960 Residents/Planner
MI	05/1966	3	90	30	189K
TX	03/1966	4	78	19.5	145K
MN	03/1966	4	74	18.5	60K
WA	03/1966	3	33	11	80K
RI	10/1957	5	13	2.6	29K

Notes: For a given year and month, we reference the project directory to see which cities had an active grant as of that point in time. The rightmost columns are ratios of the active municipalities and of those municipalities' 1960 Census populations. Bolded cells reflect caseload ratios above the HHFA caseload limit of 12 to 16 municipalities per agency planner.

Sources for the planner counts are: Gillings (1967) for Michigan, Hammer Greene Siler Associates (1969) for Texas, Minnesota Department of Business Development (1966), Washington Department of Commerce (1966) and Simha (1957) for Rhode Island.

Siler Associates (1969) notes that indeterminate delays to the start of planning work can harm the initial enthusiasm local officials had to learn about comprehensive planning. Second, the guidelines published by the HHFA indicate a requirement for state agencies is to have an active caseload of no more than 12 to 16 municipalities per planner. Documents describing Michigan and Washington's 701 experiences mention that federal authorities had warned a suspension in funds to those states until they increase their staffing numbers.

Finally, due to demand for 701 increasing nationwide, by the mid-1960s there were reports in state documents of municipality backlogs: cities that filled out the paperwork but whose applications were not processed before a year's appropriations for 701 assistance ran out. We find it plausible that states with lower capacity and higher caseloads per planner were less capable of clearing their backlogs, such that many interested municipalities could not get their applications approved before the late 1960s. Being aware of excess demand for the program, the HHFA first set additional caps on grant amounts, and then imposed a moratorium on any applications from October 1966 to the start of 1968 (Minnesota Department of Business Development, 1966).

By the late 1960s, 701 funding had expanded to many potential state and local planning authorities or government agencies, and for purposes beyond physical planning of the built environment. Additional requirements and proof of local planning capacity were put in place for the allocation of 701 funds after 1968. The extra administrative costs behind any 701 applications, consistent with a reorienting of the program towards funding regional and state planning authorities, could explain the decline in 701 assistance adopters we backed out from the imputation process in Appendix C.

D.2 Adoption rates are well explained by state agency strategies

We conducted more quantitative analyses that show how the capacity of state agencies mattered for 701 receipt. The main idea is that we check how predictive dimensions of state

agency conduct are on the adoption rate measure we use to rank states by agency capacity, $\tilde{Z}_s$. If adoption rates are driven mainly by differences in municipality characteristics between states, regressions excluding those characteristics to predict $\tilde{Z}_s$ should have poor fit. As it is unlikely we can get systematic data for the most obvious measures of capacity (e.g. the number of planners on staff at each agency), we focus on two measures that can be constructed from the 1966 project directory data.

First, we can calculate per-capita grant expenditures for each state. For each 701 grant approved in the directory, we sum the 1960 population counts for the municipalities to which the grant was applied. Figure D.1 shows mean levels of per-capita expenditures taken over the grants, further weighed by the population to which each grant was applied. Across the grants we have in the project directory, most states had per-capita expenditures of between 1.2 to 1.8 constant dollars per capita for their 701 grants. However, there are both Southern states with lower per-capita expenditures (like Alabama and North Carolina), as well as high-generosity states like Arizona.

Second, we leverage a loophole that state agencies might have exploited in the early phase of the Section 701 program. After 1961, state agencies can save paperwork and repeated filings by applying for a common grant applied to multiple neighboring municipalities, as long as their combined population is below 50,000. However, prior to 1961, the statute placed no limit on how many municipalities can be approved on a joint grant, as long as each municipality was below 25,000 residents. The degree to which state agencies packaged grants serving over 25,000 residents in sum is thus a proxy for the underlying state agency's knowledge of the policy, and its ability to use staff to coordinate joint applications across multiple municipalities.²

For each grant, before 1961, we classify if it served a group of municipalities that had over 25,000 population in 1950. For each state, we calculate the share of grants satisfying the criterion. With the two variables, we run a state-level regression where $\tilde{Z}_s$ is the outcome and the right hand side variables are the two measures, as well as their interaction.

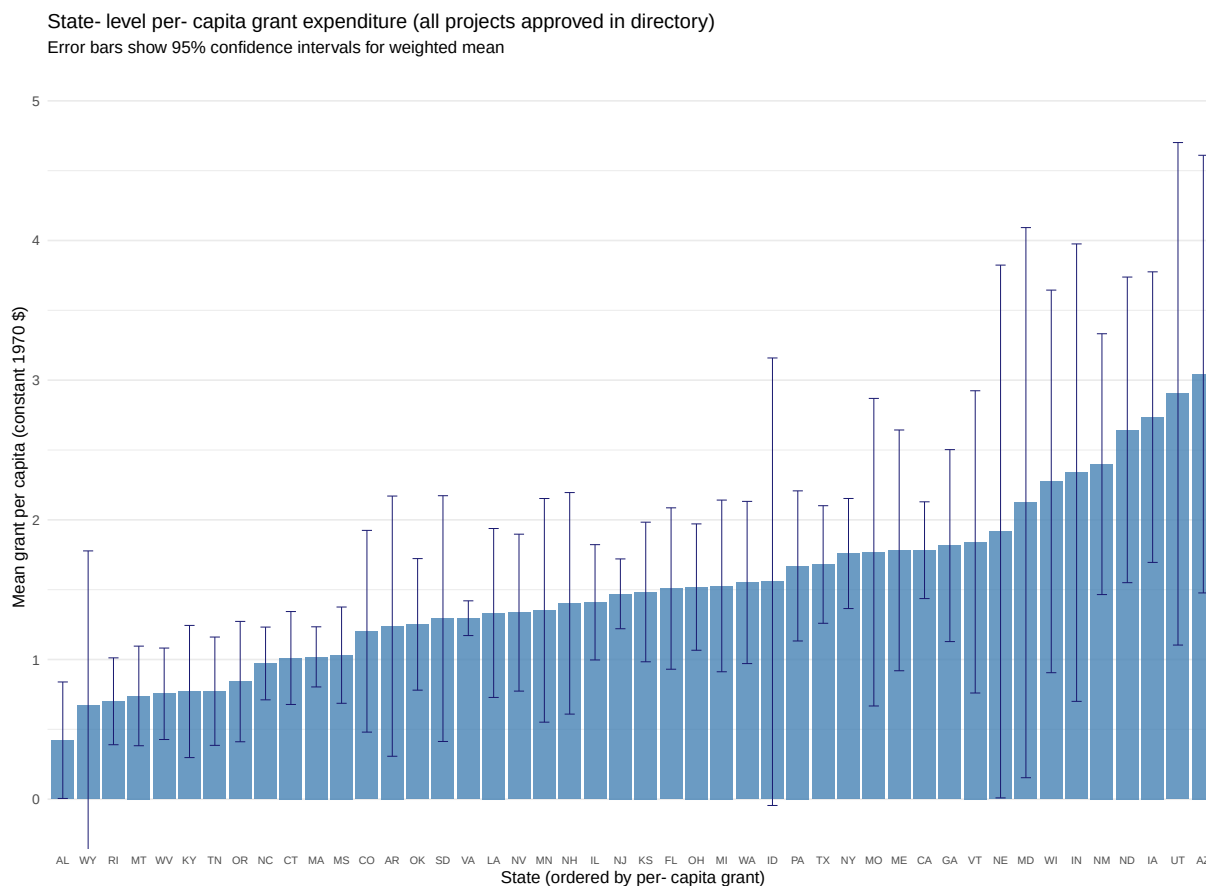
Figure D.2 shows predicted values of $\tilde{Z}_s$ from the state-agency-conduct regression, plotted against actual $\tilde{Z}_s$. The overall R^2 is 0.566, reflecting a good fit without invoking additional municipality level covariates. Outliers where the predicted rate is poor are also rare, with most predictions close to the plotted 45-degree lines. Outliers include New York and Ohio (where actual rates are lower than predicted rates) and Arkansas and Tennessee (where actual rates are higher than predicted, possibly reflecting other unobserved characteristics of high capacity with those agencies).

D.3 Comparison of empirical strategy to other research designs

D.4 Additional statistical tests

²The two state agencies that were staffed with in-house professional planners — the ones in Alabama and Rhode Island — always pooled municipal grants steadily over the year.

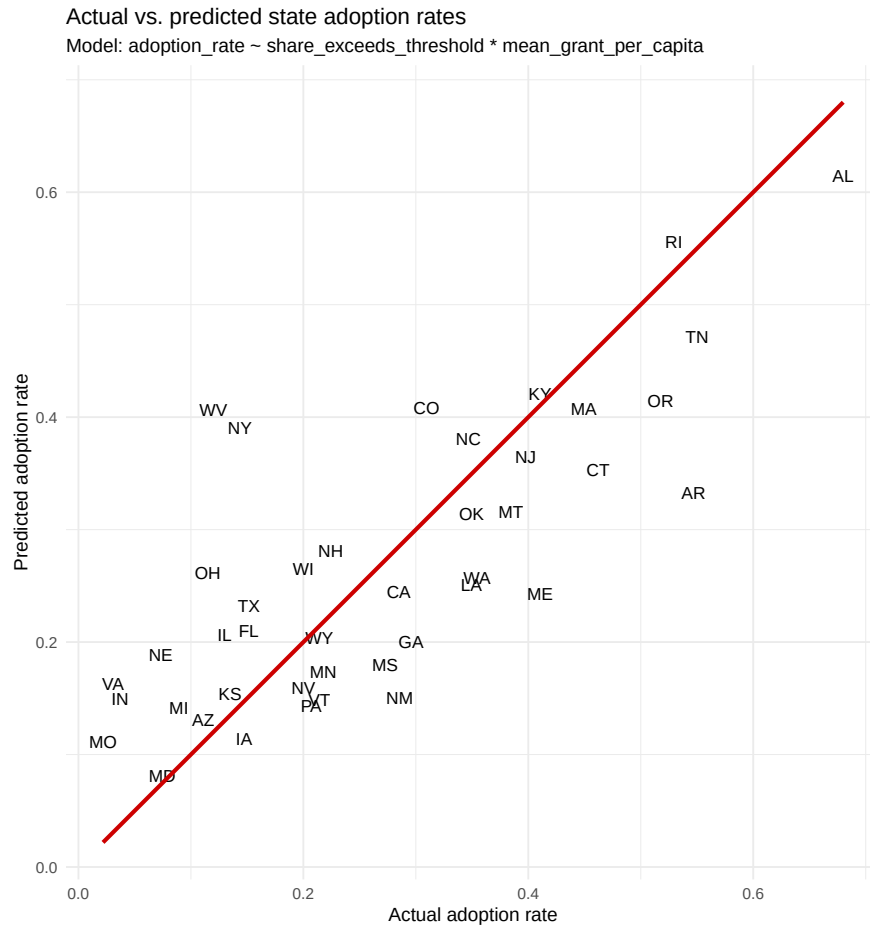
Figure D.1: Between and within-state variation in grant generosity



Notes: This figure summarizes state-level variation in per-capita grant generosity among 701 grants in the project-directory data. For each grant, per-capita spending is calculated by dividing the federal grant amount by the 1960 population of the municipalities to which the grant applied; state means are weighted by the population covered by each grant.

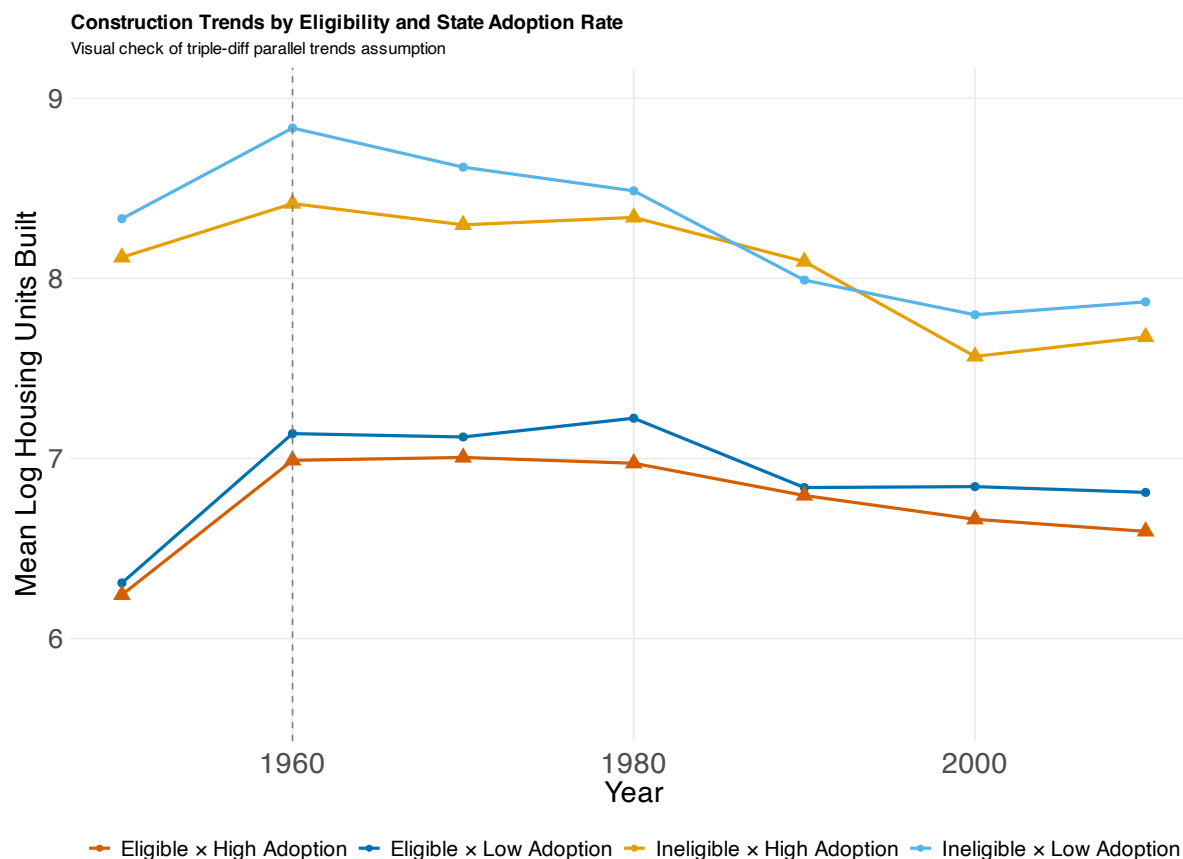
Sources: HUD/HHFA 701 project directories; Schmidt (2018) population panel data.

Figure D.2: Predicting adoption rates based on only state agency strategies



Notes: This figure plots actual state 701 adoption rates against rates predicted using only state agency strategy measures. The prediction uses state-level per-capita grant generosity, the pre-1961 share of grants that pooled municipalities with combined populations above 25,000, and their interaction. The red line is the 45-degree line.
Sources: HUD/HHFA 701 project directories; Schmidt (2018) population panel data.

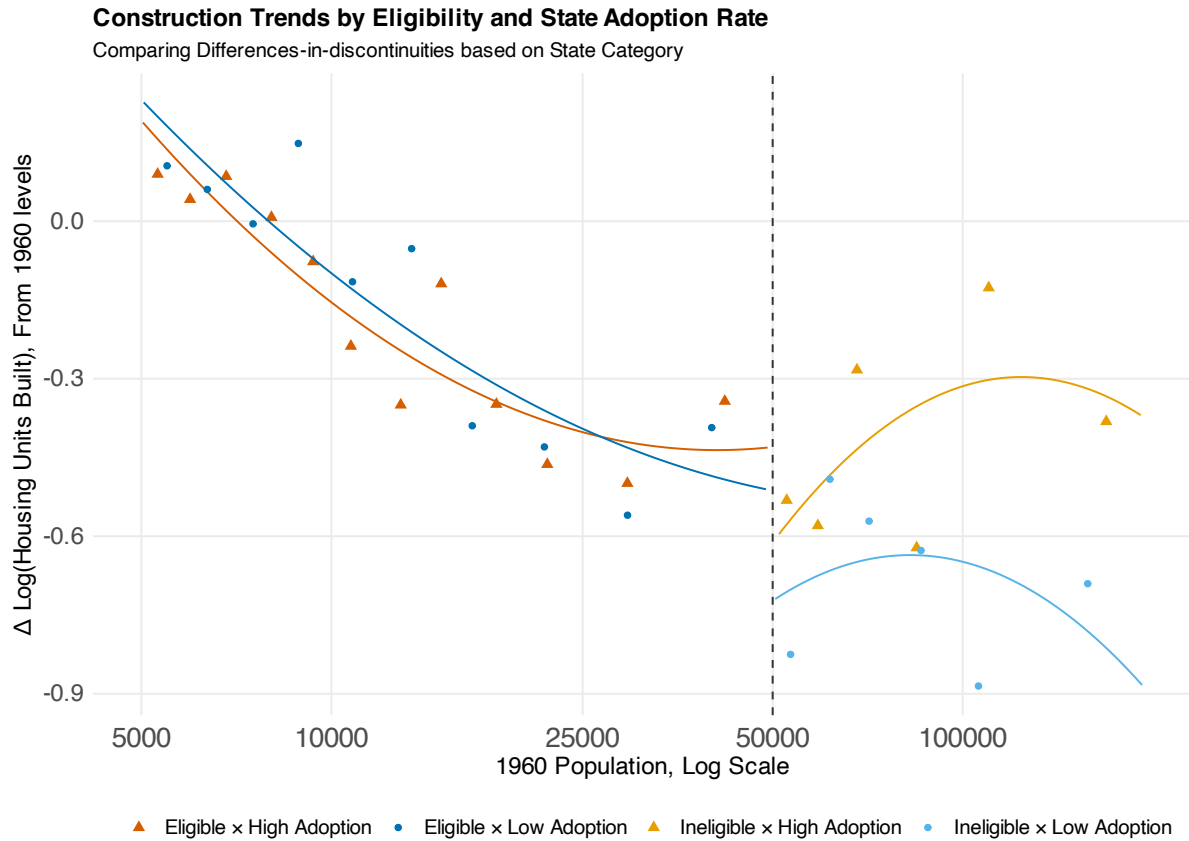
Figure D.3: Representing ID strategy as two difference-in-differences



Notes: This figure illustrates the triple-difference design as the difference between two difference-in-differences comparisons. Outcomes are plotted separately for municipalities below and above the 50,000-person eligibility threshold and for states with high versus low 701 adoption rates. The visual comparison corresponds to the design intuition developed in Section 4.2.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data.

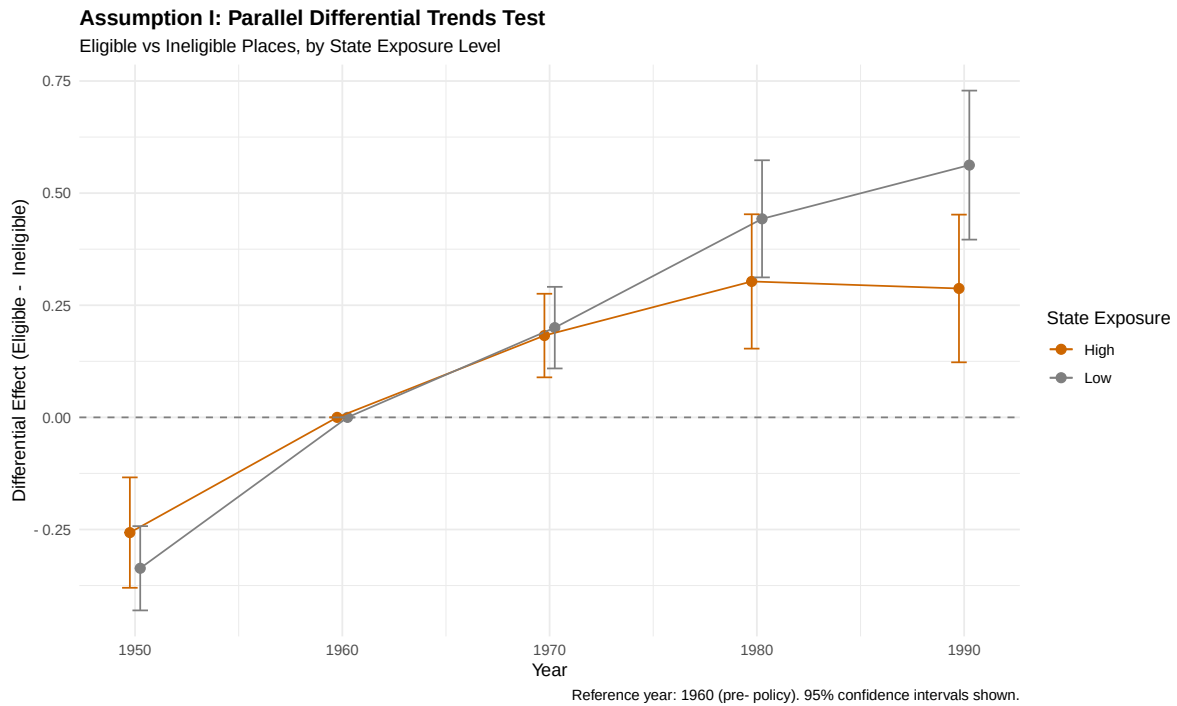
Figure D.4: Representing ID strategy as two differences-in-discontinuities



Notes: This figure illustrates the research design as a comparison of discontinuities around the 50,000-person 1960 population eligibility threshold. The design compares the eligibility discontinuity in states with higher 701 adoption rates to the corresponding discontinuity in states with lower adoption rates.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data.

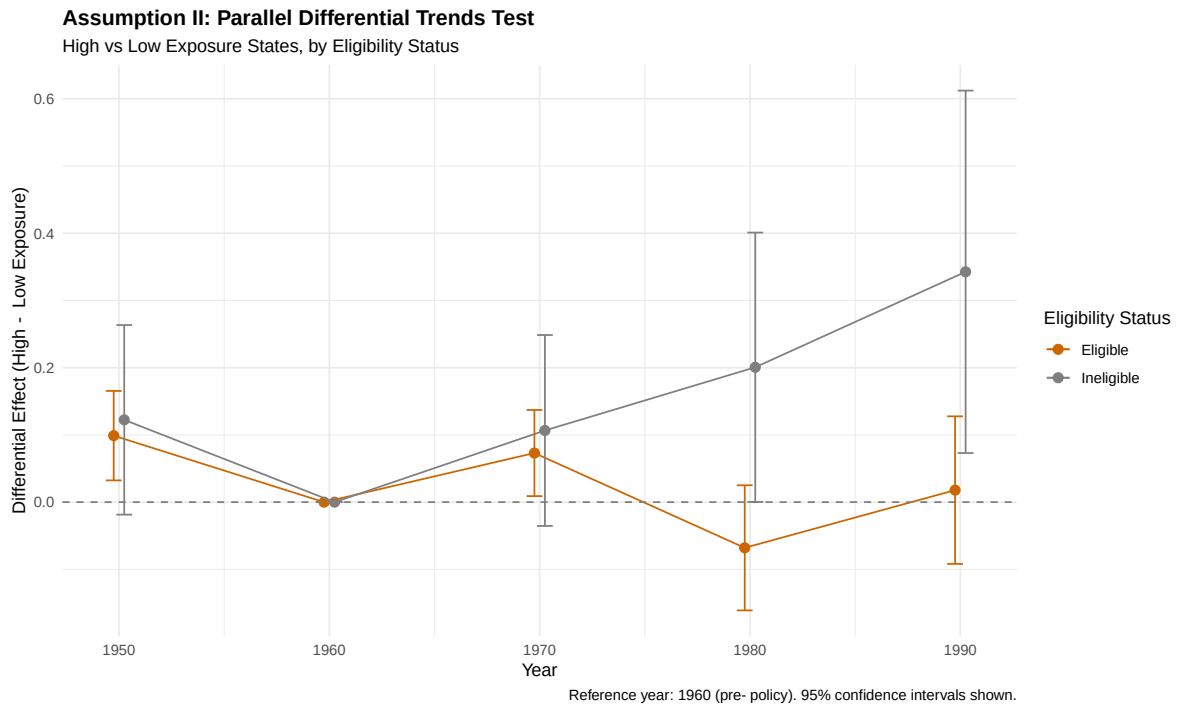
Figure D.5: Parallel trends comparisons for Lin and Peri (2025), Assumption I



Notes: This figure visualizes the pre-period comparison used to evaluate the parallel-bias condition following Lin and Peri (2025). The test compares pre-period outcome changes for municipalities below and above the 50,000-person eligibility threshold across high- and low-adoption states. Points report event-study coefficients relative to 1960; bars report 95% confidence intervals. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data.

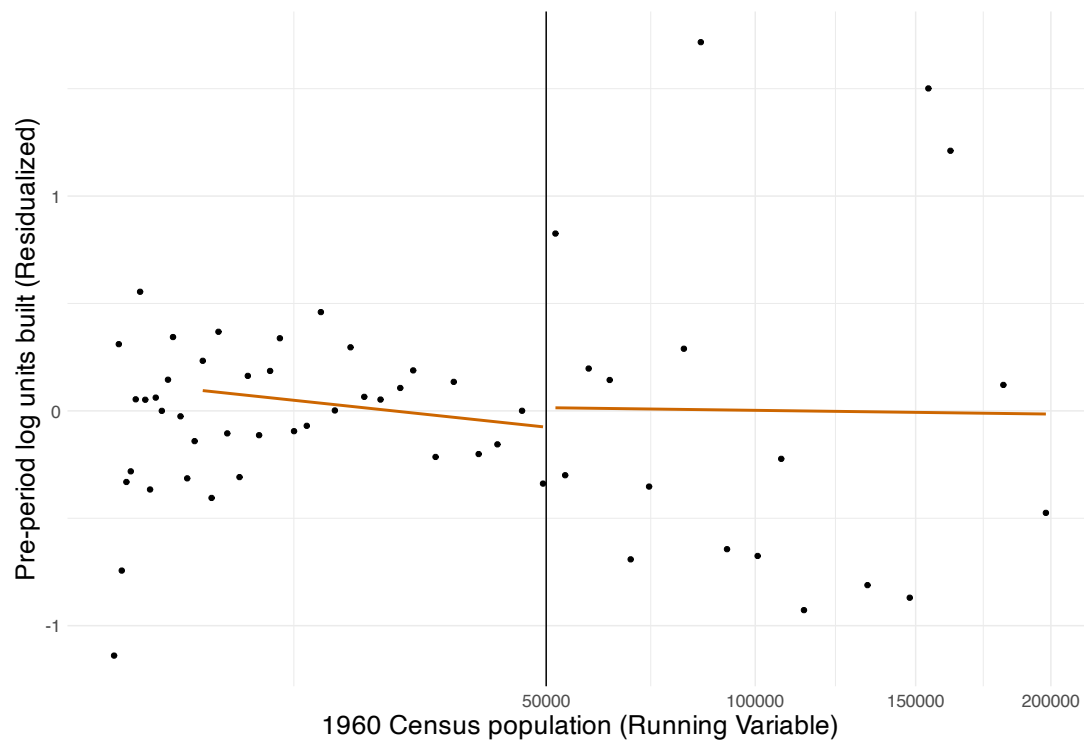
Figure D.6: Parallel trends comparisons for Lin and Peri (2025), Assumption II



Notes: This figure visualizes an alternative pre-period comparison following Lin and Peri (2025). The test separately compares pre-period outcome changes across high- and low-adoption states for municipalities on each side of the 50,000-person eligibility threshold. Points report event-study coefficients relative to 1960; bars report 95% confidence intervals. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data.

Figure D.7: Pre-period conditional independence over analysis sample thresholds



Notes: This figure plots pre-period residualized housing construction against 1960 population to assess the population window over which the conditional-independence assumption is plausible. The exercise follows the intuition of Angrist and Rokkanen (2015): the analysis sample is restricted to windows in which residualized pre-period outcomes are approximately flat around the eligibility threshold. The vertical line marks the 50,000-person eligibility threshold.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data.

D.5 Proof of Proposition in Section 4

Proof. Assumption CI directly implies, for $z, z' \in \text{supp}(\tilde{Z})$,

$$\begin{aligned} & \mathbb{E}[Y_t^0 - Y_{t-1}^0 | \tilde{Z} = z, \text{Pop}_{1960} \in (\underline{P}, 50K], X] - \mathbb{E}[Y_t^0 - Y_{t-1}^0 | \tilde{Z} = z, \text{Pop}_{1960} \in (50K, \bar{P}], X] \\ &= \\ & \mathbb{E}[Y_t^0 - Y_{t-1}^0 | \tilde{Z} = z', \text{Pop}_{1960} \in (\underline{P}, 50K], X] - \mathbb{E}[Y_t^0 - Y_{t-1}^0 | \tilde{Z} = z', \text{Pop}_{1960} \in (50K, \bar{P}], X], \end{aligned}$$

which also implies that $\mathbb{E}[Y_t^0 - Y_{t-1}^0 | \tilde{Z} = z, E_t, X] = \gamma_0 + \gamma_1 E_t$, where the analysis sample is filtered to the window $\text{Pop}_{1960} \in (\underline{P}, \bar{P}]$ and $E_t = 1$ if $\text{Pop}_{1960} \leq 50K$ and 0 otherwise.

Define a vector of covariates $\tilde{X} = [1 \quad D \times E_t \quad (1-D) \times E_t]$. The assumptions we imposed imply the following two linearity conditions:

$$\mathbb{E}[Y_t^0 - Y_{t-1}^0 | \tilde{Z}, \tilde{X}, X] = \tilde{X}' \gamma, \quad (6)$$

$$\mathbb{E}\left[\frac{Y_t - Y_t^0}{\tilde{Z}} \middle| \tilde{Z}, \tilde{X}, X\right] = \tilde{X}' \eta, \quad (7)$$

but a restricted version where the coefficients on $D \times E_t$, $(1-D) \times E_t$ equal zero in Equation 6, while coefficients on 1 is equal to 0 in Equation 7.

Using the arguments for identifying TWFE with covariates in de Chaisemartin et al. (2025), we can express:

$$\mathbb{E}[Y_t - Y_{t-1} | \tilde{Z}, \tilde{X}, X] = \tilde{X}' \gamma + \tilde{Z} \tilde{X}' \eta,$$

and a regression featuring the covariates in $\tilde{X}$ and their interactions with $\tilde{Z}$, but specifically ones whose coefficient terms in the two linearity conditions are assumed to be nonzero.

Our estimands of interest, τ_{treat} and τ_{elig} are treatment effects conditional on $D \times E_t$ and E_t , respectively. The treatment effects can be rewritten as:

$$\begin{aligned} \tau_{treat} &= \mathbb{E}\left[\tilde{Z} \times \left(\frac{Y_t - Y_t^0}{\tilde{Z}}\right) \middle| D \times E_t\right] \\ &= \mathbb{E}\left[\tilde{Z} \times \tilde{X}' \eta \middle| D \times E_t\right] \\ &= \eta^{D \times E} \mathbb{E}[\tilde{Z} | D \times E_t], \end{aligned}$$

where $\eta^{D \times E}$ reflect the coefficients for the coefficient for specifically the $D \times E_t$ coefficient.

Likewise,

$$\begin{aligned}
\tau_{elig} &= \mathbb{E} \left[\tilde{Z} \times \left(\frac{Y_t - Y_t^0}{\tilde{Z}} \right) \middle| E \right] \\
&= \mathbb{E} \left[\mathbb{E}[\tilde{Z} \times \tilde{X}'\eta \mid D] \middle| E \right] \\
&= \begin{bmatrix} \eta^{D \times E} \mathbb{P}[D = 1 \mid E] & \eta^{(1-D) \times E} \mathbb{P}[D = 0 \mid E] \end{bmatrix} \begin{bmatrix} \mathbb{E}[\tilde{Z} \mid D = 1, E] \\ \mathbb{E}[\tilde{Z} \mid D = 0, E] \end{bmatrix}.
\end{aligned}$$

□

We briefly discuss the plausibility of the assumption that there is no “intercept” coefficient in Equation 7. Our preferred interpretation of that assumption is that for the smallest values of $\tilde{Z}$, the coefficient being zero implies 701-eligible municipalities after the policy rollout did not alter their growth trajectories, nor cause spillover effects on larger ineligible cities.

In our context, because we have near-zero levels of $\tilde{Z}$ for certain states, those states are analogous to the “quasi-untreated group” invoked by de Chaisemartin et al. (2025) as necessary for effect identification. It is then unlikely that states where 701 assistance was rarely applied could generate intercept effects, and the assumption on no spillovers to ineligible cities is no stronger than what is implied by invoking Assumption CI.

E Details on LLM parsing of developments awaiting permits

We detail the procedure we used to code a novel sample of newspaper articles that describe specific developments that had to be approved for a building permit or land entitlement. The source of our local newspaper data is the Newsbank Access World News database, which is licensed for non-commercial use to New York University. We processed newspapers between the years of 1983 to 2009 throughout the United States.

We first retrieve a sample of articles specifically about developments that satisfy two qualities: the article mentions it is not yet approved or permitted, and the article mentions some form of opposition to a developer’s plans. To do so, we used the Newsbank platform to search for articles with the following three terms in the article body: “housing units”, “developer*” and “neighbor* oppos*”. The asterisks are wildcard terms, so i.e. both “developer” and “developers” would be searched for in this search. We also require the article to either have the term “approv*” or “permit.”

About 4,500 articles are returned from the search, a sample that is biased towards recent years due to data availability. We undersample articles in recent years while processing nearly all available articles in early years. The sample thus includes around 180 articles prior to 1990, up to 360 articles in the 2003-5 3-year period.

For the final sample of around 2,000 articles, we process each article through three separate LLM prompts to verify their relevance and extract details of the permitting process. We leverage the gpt-5-mini model from OpenAI and iteratively execute the prompts through the DocETL data processing platform. The tasks we prompt the LLM to do are either retrieval of specific terms from the text, or annotation tasks that ask if the article describes a certain trait or practice; the LLM then returns a binary yes/no answer and potentially an “N/A”. Our process thus closely resembles the tasks instructed to the LLM in Bartik, Gupta and Milo (2024) and Djourelouva et al. (2024). To ensure accuracy, we also execute our prompts in a “few-shot” manner by including examples of the tasks at hand.

We produce the three prompts verbatim below. The first prompt works to filter the article out of later processing, if the LLM decides the article does not concern a residential development at all. The second prompt extracts characteristics of the development, while the third prompt asks for any description of specific processes. Variables embedded in curly brackets reflect variables referenced through the DocETL interface.

Relevance prompt:

Analyze the following news article:

Title: "{{ input.title }}"

Content: "{{ input.body }}"

Determine if this article is relevant based on the following criteria:

Does the article concern the opposition to approving, or process of approving, a residential building in a city?

Some examples of articles that do not concern the approval process:

(1) An article that just describes the character of a city or neighborhood.

(2) An article about a developer's biography, without reference to a specific building under the approval/entitlement process.

Respond with 'true' if the article satisfies the criteria, otherwise respond with 'false'.

Project characteristics prompt:

Analyze the following news articles that involve some sort of local government body approving or opposing residential buildings:

Title: "{{ input.title }}"

Content: "{{ input.body }}"

Most of each article should refer to the city in which a building is to be built. Another possible case is if the local government unit is a County, in which case there should be reference to bodies named, e.g. the "county planning commission."

DO NOT confuse a reference to the city or county with a reference to an area or neighborhood. If you cannot find an unambiguous reference to the city in question, check the field in JSON: `input.city` and that value where it is available.

DO NOT include references to projects unrelated to residential buildings, such as malls, streets, infrastructure and so forth.

Respond with blank or NA values if information is unavailable.

With that knowledge, answer the following questions for every building reported on in the article:

(1) In which municipality or general local government unit did the reported event takes place? If the local body is a county, always add "County" to the end of the field. Otherwise, ignore indicators like "City of," "Town of," etc.

(2) What was the originally proposed number of housing units in the building which might be approved? This column should consist of a single number.

(3) Only if the article discusses a final number of housing units in question: what is the agreed upon number of housing units that are permitted in the approved project?

Process characteristics prompt:

Analyze the following news articles that involve some sort of local government body approving or opposing residential buildings:

Title: "{{ input.title }}"

Content: "{{ input.body }}"

Most of each article should refer to the city in which a building is to be built. Another possible case is if the local government unit is a County, in which case there should be reference to bodies named, e.g. the "county planning commission."

We will ask you about citizen interest groups, examples of which include "neighbors" and "neighborhood associations." We will also ask you about "local planning staff", examples of which include "city planners," "city planning commission" or any governmental body other than the city council voting on approving a project.

By "low-income housing", we mean housing units deed restricted to be sold at below market rents. Articles will generally refer to a specific number or share of units as allocated to "low-income residents/households," and these projects may also be mentioned as being financed by housing tax credits at the federal or state/local agency level.

Respond with blank or NA values if information is unavailable.

With that knowledge, answer the set of binary questions for every building reported on in the article, where 1=Yes, 0=No;

- (1) Are citizen interest groups quoted to have spoken out in opposition of the project?
- (2) Have local planning staff, up to now, delayed or rejected the project?
- (3) Has the residential building reduced in size as part of deliberations to get a permit to build?
- (4) Has the developer of the building offered to provide additional community benefits, or impact fees?
- (5) Are a share of the building's housing unit intended to provide low-income housing?
- (6) Is the entirety of the building intended to provide low-income housing?

Mapping of prompt questions to outcomes. As with other text-based datasets in this Appendix, we use only the developments whose municipality (from Question (1) of the project prompt) can be fuzzy matched to a directory of zoning jurisdictions. In our analysis, the distribution of development unit counts shown in Appendix Figure B.11 reflect the output of Question (2) of the project prompt.

The main value capture outcome we analyze — the presence of additional community benefits or impact fees — comes from Question (4) of the process prompt. We have manually checked 11 developments classified as "1" by the LLM, noting that all but one had an article that described a common value capture practice. Examples of correctly identified practices include developer commitments to finance any roads or infrastructure around the development; provision of public amenities, like sound barriers next to the highway and a community center; in-lieu fees to support the maintenance of affordable housing in the municipality; and leaving up to half of the acquired parcel as public open space.

Table 7 also reports two other outcomes. The low-income housing outcome maps to Question (5) in the process prompt, while the local staff opposition outcome maps to Question (2) in the process prompt. We have run our cross-sectional design on the opposition outcomes related to Questions (1), (3) and (6), all of which output null results similar to the local staff opposition outcome.

F Details of robustness checks

Robustness to sample and population threshold selection. Appendix Table A.6 produces robustness checks using different ways of cutting off the maximum and minimum 1960 populations in the analysis sample. In our triple differences design, such sample adjustments are analogous to changing the local bandwidth over which outcome functions are estimated in a regression discontinuity design (Cattaneo and Titiunik, 2022). We note that a specification where no population thresholds are used produces more negative effect estimates on housing supply. A specification with a smaller population range (1960), from 7,500 to 160,000 residents, yields similar negative effect estimates to those in Table 2, but increases the standard errors due to a smaller sample size.³

In contrast to the first two panels of Appendix Table A.6, which vary how much we include data away from the discontinuity, the last panel of the table conducts a “donut” robustness check close to the discontinuity. When we drop municipalities whose 1960 populations were between 40,000 to 60,000, supply effects are more negative across all specifications. Finally, Appendix Table A.7 conducts a placebo regression discontinuity at a nearby population level not used as a threshold in 701 assistance. In a setup similar to Aaronson, Hartley and Mazumder (2021), we run the design for municipalities with populations below 20,000 and those above 20,000 but below 50,000. For our supply and price outcomes, we find point estimates close to zero, with no significant effects at the 90% confidence level.

Post-1960s state divergence. A competing mechanism for our results is that states whose agencies were more capable of distributing 701 assistance funds may also have had a greater interest in passage of additional planning legislation. That increased interest may have been endogenous to the states’ economic structure prior to the 1960s: for example, if the state had large manufacturing sectors that induced pollution and environmental disamenities. While empirical work exists on the housing supply consequences of green belts and regional land-use restrictions (Yu (2019); Koster (2024)), those effects operate separately from our proposed channel of 701 assistance, changing perceptions about growth among local interest groups.

We verify that our results are robust to a variety of additional state-level controls marking differences in states’ industrial composition and planning laws. In Table F.1, we add two predetermined state-level controls: 1940 manufacturer share and Black resident share in incorporated municipalities, tabulated from the full count data in Section 3.3. In Table F.2, we control explicitly for the adoption of state environmental policy acts (SEPA), implementing environmental review laws that have been leveraged to delay and cancel residential development (Elmendorf and Nall, 2024). Furthermore, we also conduct an analysis excluding the three states whose SEPA allow for delays and remedies for non-compliant private-sector projects: California, New York, and Minnesota.

In the tables we listed, we find minimal changes in the estimated coefficients for housing

³As discussed in Section 4.3, the identifying variation in our research design is not local variation around the 50,000 population eligibility cutoff, but variation in state agency capacity that is plausibly exogenous to differential trends between small and large municipalities. In this context, one interpretation of the role of standard checks for an RDD is to ensure that our results are not driven by outlier values clustered around specific values of the 1960 population running variable.

supply, the cumulated apartment share, and median values. Standard errors are generally larger, but the least significant robustness result for the supply outcome maintains 90% statistical significance.

In addition, Table E3 reports the results of a saturated regression that includes predetermined state-level controls, as well as multiple indices of state planning law restrictiveness from the American Planning Association’s 2022 Survey of State Planning Laws. The survey controls include binary variables for state requirements on local planning practices, as well as a three-level index for how comprehensive state laws are for incentivizing “community resilience.” When all the controls are included, we find more negative results for housing supply and more precise estimates for all our supply and price outcomes.

Land annexation as a competing mechanism. Changes in how incorporated 701-adopting municipalities annex development around it could impact our housing supply effects in a mechanical way. The concern would arise if, after 701 assistance, municipalities annex less unincorporated territory than ones that did not receive assistance. In that case, even if the built environment and density of the two types of municipalities were the same, we would see faster log unit growth in 701-ineligible municipalities than in 701-eligible municipalities.

Ruling out this mechanism is complicated by a lack of panel data on U.S. municipality land areas that cover our analysis period and have nationwide coverage. We proceed in two parts and leverage the best land area data that are currently digitized. First, we calculate land areas for incorporated municipalities in the 1980 and 2010 TIGER files, which are comprehensive nationally. Next, we retrieve land areas prior to 1980 recorded in the CCDB collection, which is an unbalanced panel for larger municipalities with populations of over 25,000.

The two sets of results are presented jointly in Appendix Figure E1, where we apply the event-study design of Equation 4 to the data. The triple-difference estimates on annexation changes from 1980 to 2010 are close to zero. Results on pre-1980 data, as measured on the smaller CCDB sample, are not indicative of a negative treatment effect on the annexation outcome. We find, if anything, a positive treatment effect that suggests larger 701-eligible municipalities annexed more land.⁴

⁴We have also tried a specification that includes only municipalities with more than 25,000 residents as of 1960, which further balances the post-1980 data with the CCDB data. In it, we detect insignificant negative treatment effects on the land area outcome, due to wide confidence intervals.

Table F.1: Primary outcome results controlling for workforce confounders

	(1)	(2)	(3)	(4)	(5)	(6)	
	Outcomes - Restricted w/ state-level workforce composition						
	(Log) New Units	% Units (Cumul. to 2010)	Apt (Cu- to 2010)	% SF Units (Cumul. to 2010)	(Log) Med. Home Val- ues	Moderate Owned Unit Shares	MLS Re- quirement Restrictive- ness
Coefficient	-0.631** (0.309)	-7.942** (3.523)		8.843** (3.590)	0.313** (0.132)	-14.189* (7.486)	5.085** (2.000)
Observations	10,654	10,654		10,654	9,746	8,906	12,145
R ² (within)	0.043	0.031		0.021	0.070	0.069	0.022

Significance levels: * = 10%; ** = 5%; *** = 1%.

Notes: This table reports robustness estimates that control for predetermined state-level industrial and demographic differences. Columns report the outcome listed in the column heading. Coefficients are from variants of the triple-difference specification in Equation 2, estimated on the restricted sample of municipalities with 1960 populations between 5,000 and 200,000. The specification adds state-level 1940 manufacturing-worker share and Black household-head share controls. Standard errors are clustered at the municipality level.

Sources: IPUMS 1940 Full-Count Census tabulations (Ruggles et al., 2021); IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; Cui (2024) lot-size restrictiveness measures; and transportation access controls from Atack (2017) and Weiwu (2024).

Table F.2: Primary outcome results controlling for state environmental protection

	(1)	(2)	(3)	(4)	(5)	(6)	
	Outcomes - Restricted w/ SEPA adoption dummy						
	(Log) New Units	% Units (Cumul. 2010)	Apt (Cu- to 2010)	% SF Units (Cumul. to 2010)	(Log) Med. Home Val- ues	Moderate Owned Unit Shares	MLS Re- quirement Restrictive- ness
Coefficient	-0.641** (0.310)	-7.888** (3.590)		9.161** (3.635)	0.255** (0.127)	-12.270 (7.568)	6.163*** (1.985)
Observations	10,654	10,654		10,654	9,746	8,906	12,145
R ² (within)	0.041	0.029		0.018	0.075	0.068	0.018
	Outcomes - Restricted w/out SEPA states						
	(Log) New Units	% Units (Cumul. 2010)	Apt (Cu- to 2010)	% SF Units (Cumul. to 2010)	(Log) Med. Home Val- ues	Moderate Owned Unit Shares	MLS Re- quirement Restrictive- ness
Coefficient	-0.604* (0.312)	-9.130*** (3.508)		6.961* (3.743)	0.237* (0.121)	-17.525** (7.546)	3.956** (2.010)
Observations	8,400	8,400		8,400	7,716	7,054	9,569
R ² (within)	0.048	0.045		0.043	0.086	0.085	0.025

Significance levels: * = 10%; ** = 5%; *** = 1%.

Notes: This table tests whether the primary results are explained by state environmental policy acts (SEPA). Columns report the outcome listed in the column heading. Coefficients are from variants of the triple-difference specification in Equation 2. The first panel adds a SEPA adoption control; the second panel excludes California, New York, and Minnesota. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; Cui (2024) lot-size restrictiveness measures; authors' coding of state environmental policy act adoption dates; and transportation access controls from Attack (2017) and Weiwu (2024).

Table F.3: Primary outcome results controlling for statewide planning legislation

	(1)	(2)	(3)	(4)	(5)	(6)	
	Outcomes - Restricted w/ planning index controls						
	(Log) New Units	% Units (Cumul. 2010)	Apt (Cu- to 2010)	% SF Units (Cumul. to 2010)	(Log) Med. Home Val- ues	Moderate Owned Unit Shares	MLS Re- quirement Restrictive- ness
Coefficient	-0.780** (0.306)	-9.232*** (3.496)	10.038*** (3.555)	0.345*** (0.130)	-17.206** (7.484)	4.567** (1.927)	
Observations	10,654	10,654	10,654	9,746	8,906	12,145	
R ² (within)	0.059	0.047	0.033	0.077	0.078	0.033	

Significance levels: * = 10%; ** = 5%; *** = 1%.

Notes: This table reports robustness estimates that control for broader statewide planning legislation. Columns report the outcome listed in the column heading. Coefficients are from variants of the triple-difference specification in Equation 2, estimated on the restricted sample of municipalities with 1960 populations between 5,000 and 200,000. The specification adds predetermined state-level industrial and demographic controls and state planning-law controls for planning mandates, vertical consistency requirements, and community-resilience planning. Standard errors are clustered at the municipality level.

Sources: IPUMS 1940 Full-Count Census tabulations (Ruggles et al., 2021); IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; Cui (2024) lot-size restrictiveness measures; American Planning Association's 2022 Survey of State Planning Laws; and transportation access controls from Attack (2017) and Weiwu (2024).

Table F.4: Primary outcome results controlling for urban renewal intensity

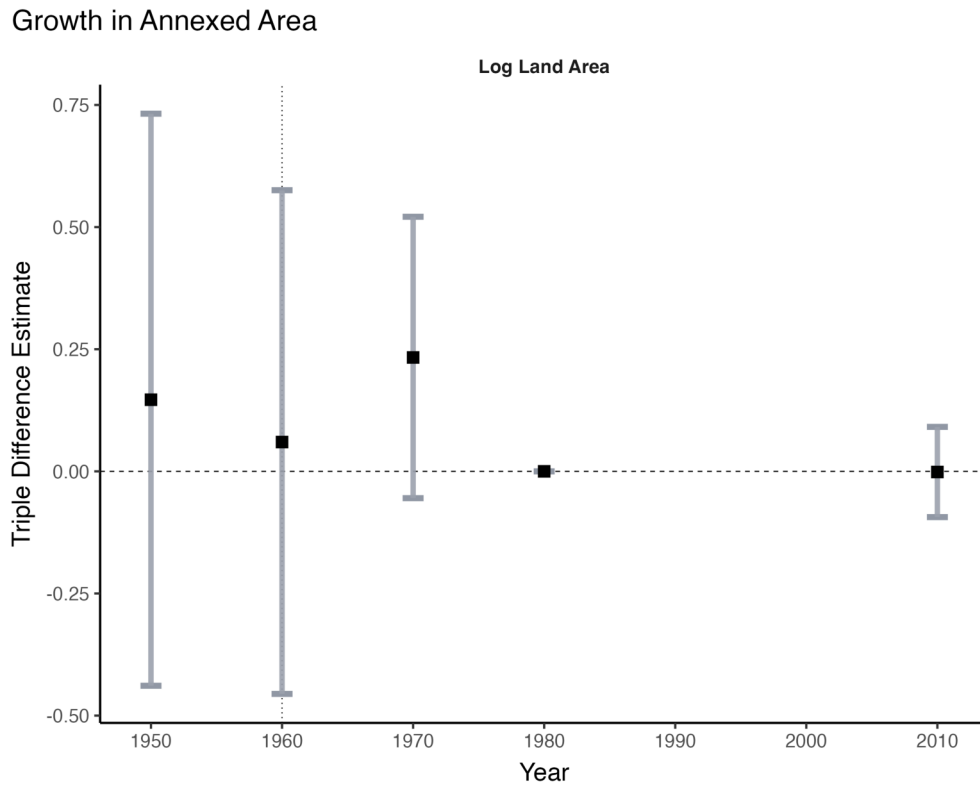
	(1)	(2)	(3)	(4)	(5)	(6)	
	Outcomes - Restricted w/ urban renewal intensity controls						
	(Log) New Units	% Units (Cumul. to 2010)	Apt (Cu- to 2010)	% SF Units (Cumul. to 2010)	(Log) Med. Home Val- ues	Moderate Owned Unit Shares	MLS Re- quirement Restrictive- ness
Coefficient	-0.646** (0.309)	-8.739** (3.465)		10.180*** (3.592)	0.291** (0.125)	-17.373** (7.469)	4.656** (2.080)
Observations	10,654	10,654		10,654	9,746	8,906	12,145
R ² (within)	0.063	0.059		0.045	0.082	0.075	0.033

Significance levels: * = 10%; ** = 5%; *** = 1%.

Notes: This table reports robustness estimates that control for state-level urban renewal and slum-clearance intensity. Columns report the outcome listed in the column heading. Coefficients are from variants of the triple-difference specification in Equation 2, estimated on the restricted sample of municipalities with 1960 populations between 5,000 and 200,000. The specification adds state-level controls for 1970 urban renewal allocations, 1970 slum-clearance allocations, urban renewal enabling-legislation timing, and other state-level planning and demographic controls. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; Cui (2024) lot-size restrictiveness measures; authors' tabulations of state-level urban renewal and slum-clearance data; and transportation access controls from Atack (2017) and Weiwu (2024).

Figure F.1: Triple difference event studies for available annexation outcome

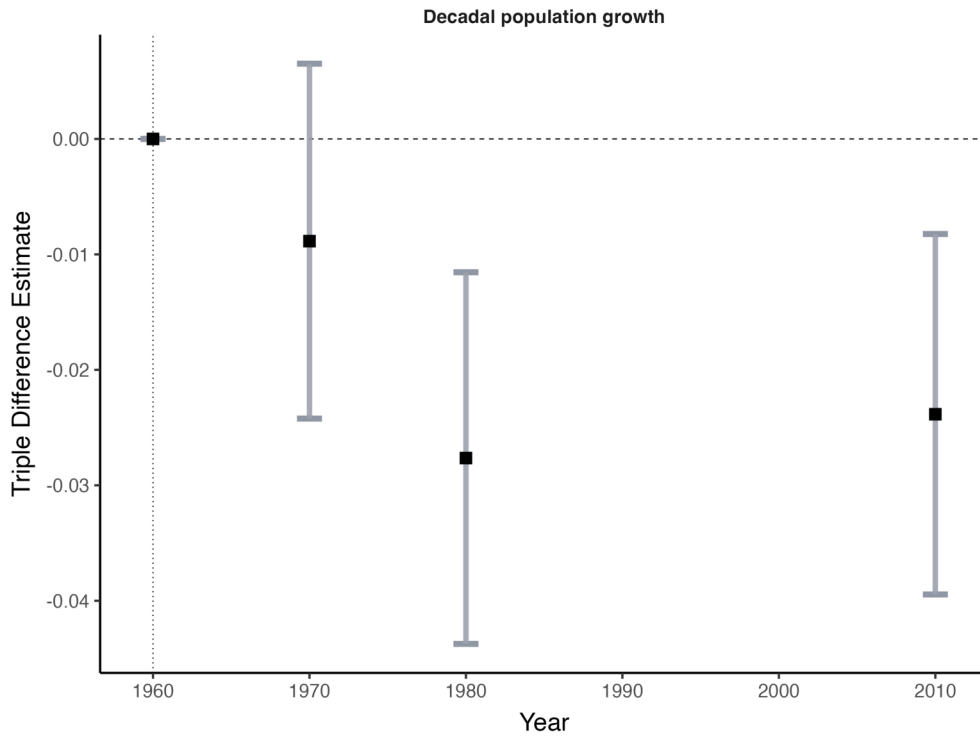


Notes: This figure plots coefficients from the event-study version of the triple-difference design in Equation 4 for available municipality land-area outcomes. Estimates use the restricted sample of municipalities with 1960 populations between 5,000 and 200,000 when data are available. Points report event-time coefficients; bars report 95% confidence intervals. Standard errors are clustered at the municipality level.

Sources: Census place boundary files; City and County Data Books; HUD/HHFA 701 project directories; Schmidt (2018) population panel data; and transportation access controls from Atack (2017) and Weiwu (2024).

Figure F2: Triple difference event studies for Census population outcome

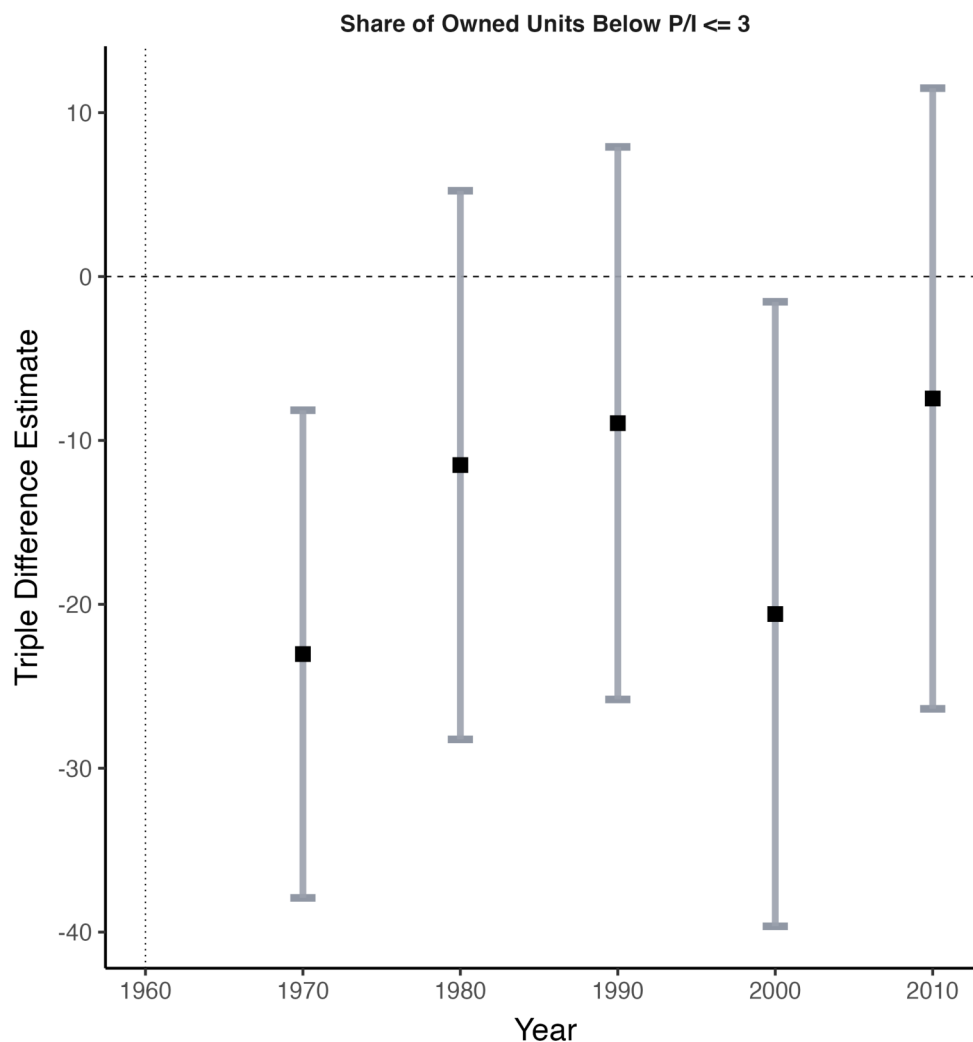
Slowdown in Population Growth



Notes: This figure plots coefficients from the event-study version of the triple-difference design in Equation 4 for Census population. Estimates use the restricted sample of municipalities with 1960 populations between 5,000 and 200,000. Points report event-time coefficients; bars report 95% confidence intervals. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; and transportation access controls from Atack (2017) and Weiwu (2024).

Figure E.3: Triple difference event studies for price-to-income affordability measure



Notes: This figure plots coefficients from the event-study version of the triple-difference design in Equation 4 for a price-to-income affordability measure. Estimates use the restricted sample of municipalities with 1960 populations between 5,000 and 200,000. Points report event-time coefficients; bars report 95% confidence intervals. Standard errors are clustered at the municipality level.

Sources: IPUMS NHGIS tables (Schroeder et al., 2025); HUD/HHFA 701 project directories; Schmidt (2018) population panel data; and transportation access controls from Atack (2017) and Weiwu (2024).